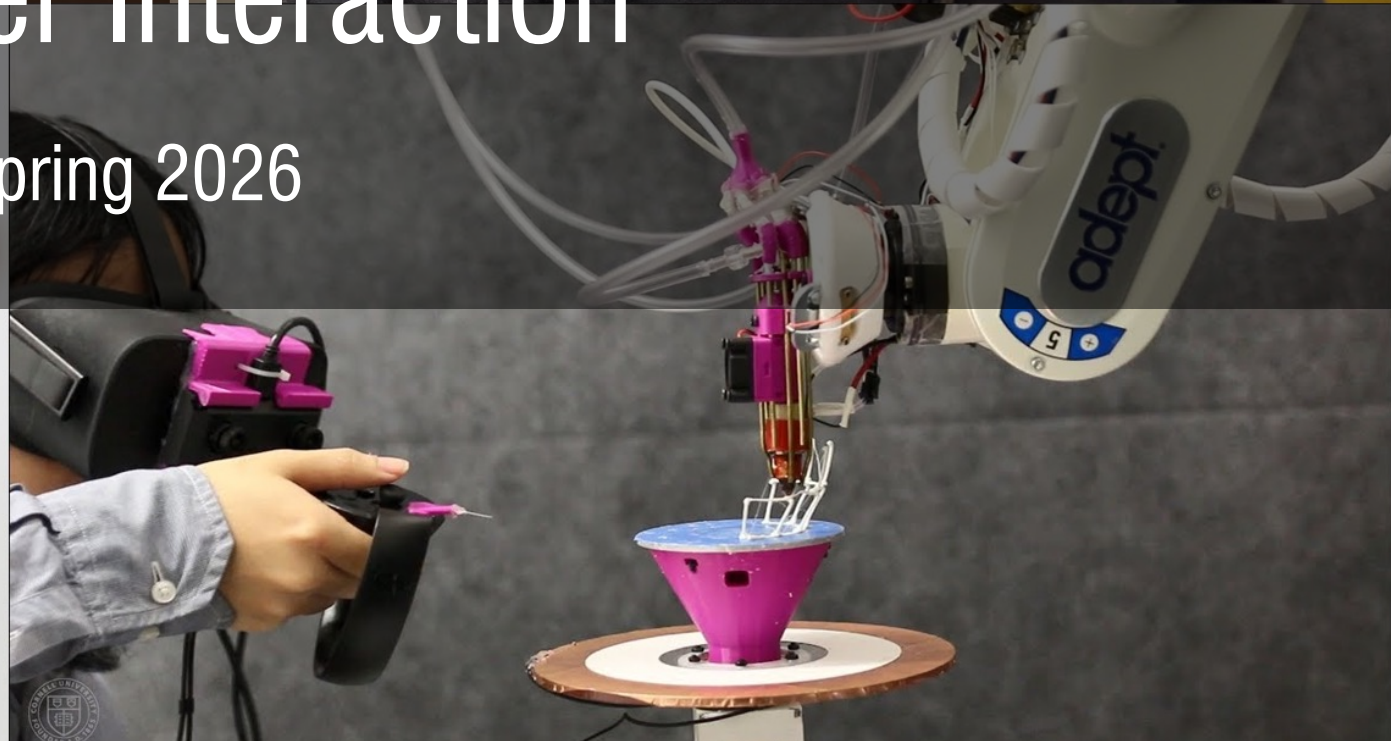




Interactive Technology for Human-Computer Interaction

Huaishu Peng | UMD CS | Spring 2026



TA: Zeyu Yan

zeyuy@umd.edu

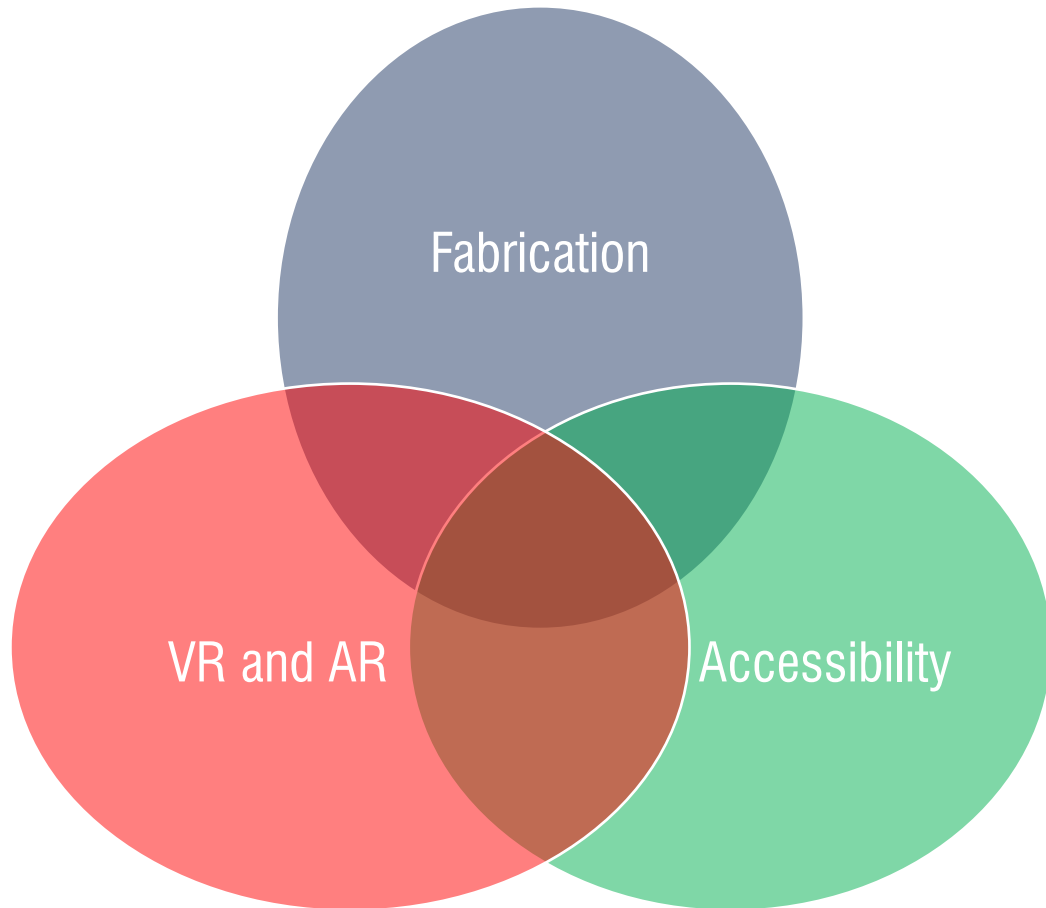
PhD Student, CS

He works in the area of Human Computer Interaction (HCI) within the field of Digital Fabrication, Tangible User Interface, Accessibility, and Haptic Interaction. His recent work revolves around digital fabrication techniques that supports sustainable printed circuit board (PCB) prototyping.

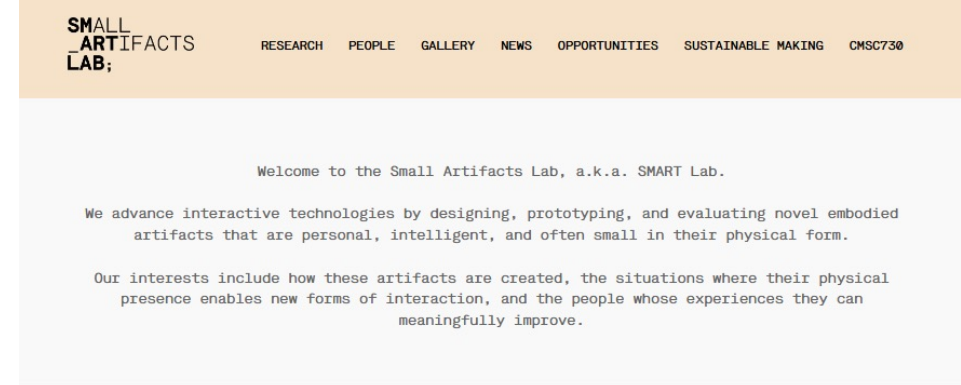


Huaishu Peng

Assistant Professor | UMD CS/UMIACS/HCIL
huaishu@umd.edu

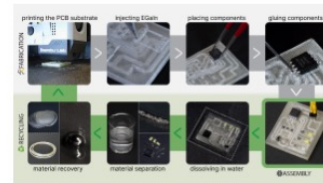


<https://smartlab.cs.umd.edu/>

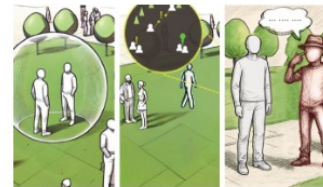


Publications

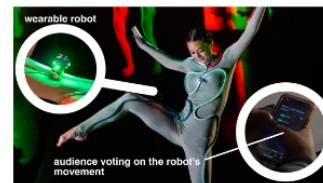
UIST 2025 | UIST Best Paper Award
DissolvPCB: Fully Recyclable 3D-Printed Electronics with Liquid Metal Conductors and PVA Substrates
Zoyu Yan, Suhwan Hong, Josiah Hestor, Tingyu Cheng, Huaishu Peng
[DOI](#) [PDF](#) [GITHUB](#)



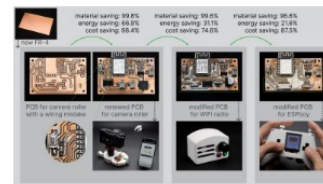
CSCW 2025
Verisimilitude as Boon and Bane: How People Initiate Opportunistic Interactions at Professional Events in Social VR
Victoria Chang, Caro Williams-Pierco, Huaishu Peng, Ge Gao
[DOI](#) [PDF](#)



DIS 2025
Cybernetic Marionette: Channeling Collective Agency Through a Wearable Robot in a Live Dancer-Robot Duet
Anup Sathya, Jiasheng Li, Zoyu Yan, Adriano Fang, Bill Kules, Jonathan David Martin, Huaishu Peng
[DOI](#) [PDF](#) [GITHUB](#)



CHI 2025 | CHI Best Paper Honorable Mentions Award
PCB Renewal: Iterative Reuse of PCB Substrates for Sustainable Electronic Making
Zoyu Yan, Advait Vartak, Jiasheng Li, Zining Zhang, Huaishu Peng
[DOI](#) [PDF](#) [GITHUB](#)



CHI 2025
Make Making Sustainable: Exploring Sustainability Practices, Challenges, and Opportunities in Making Activities
Zoyu Yan, Mrunal Dhaygude, Huaishu Peng
[DOI](#) [PDF](#)



RoMA: Interactive Fabrication with Augmented Reality and a Robotic 3D Printer

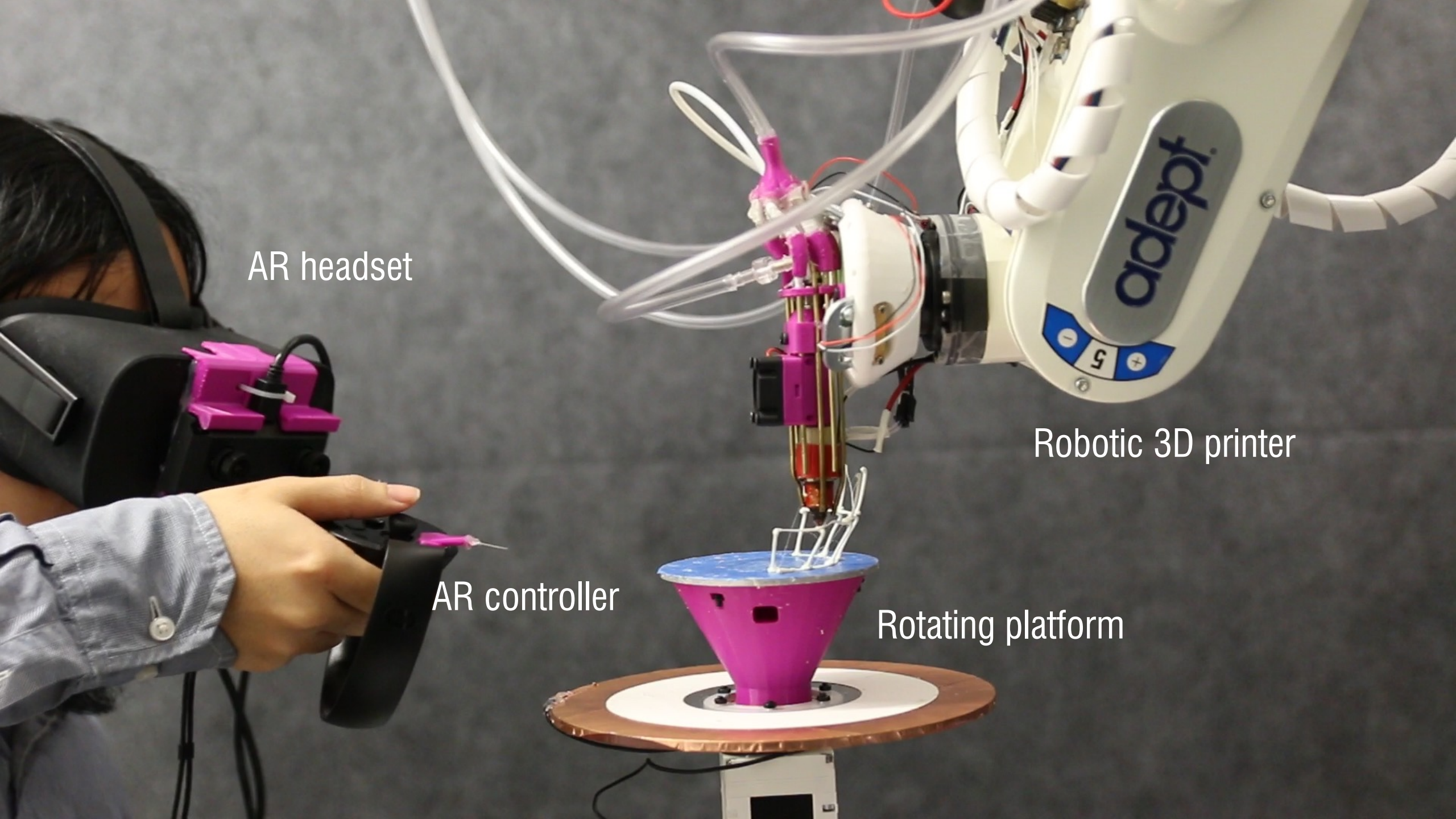
CHI 2018

Huaishu Peng | Jimmy Briggs | Cheng-Yao Wang | Kevin Guo | Joseph Kider | Stefanie Mueller | Patrick Baudisch | François Guimbretière



Cornell University



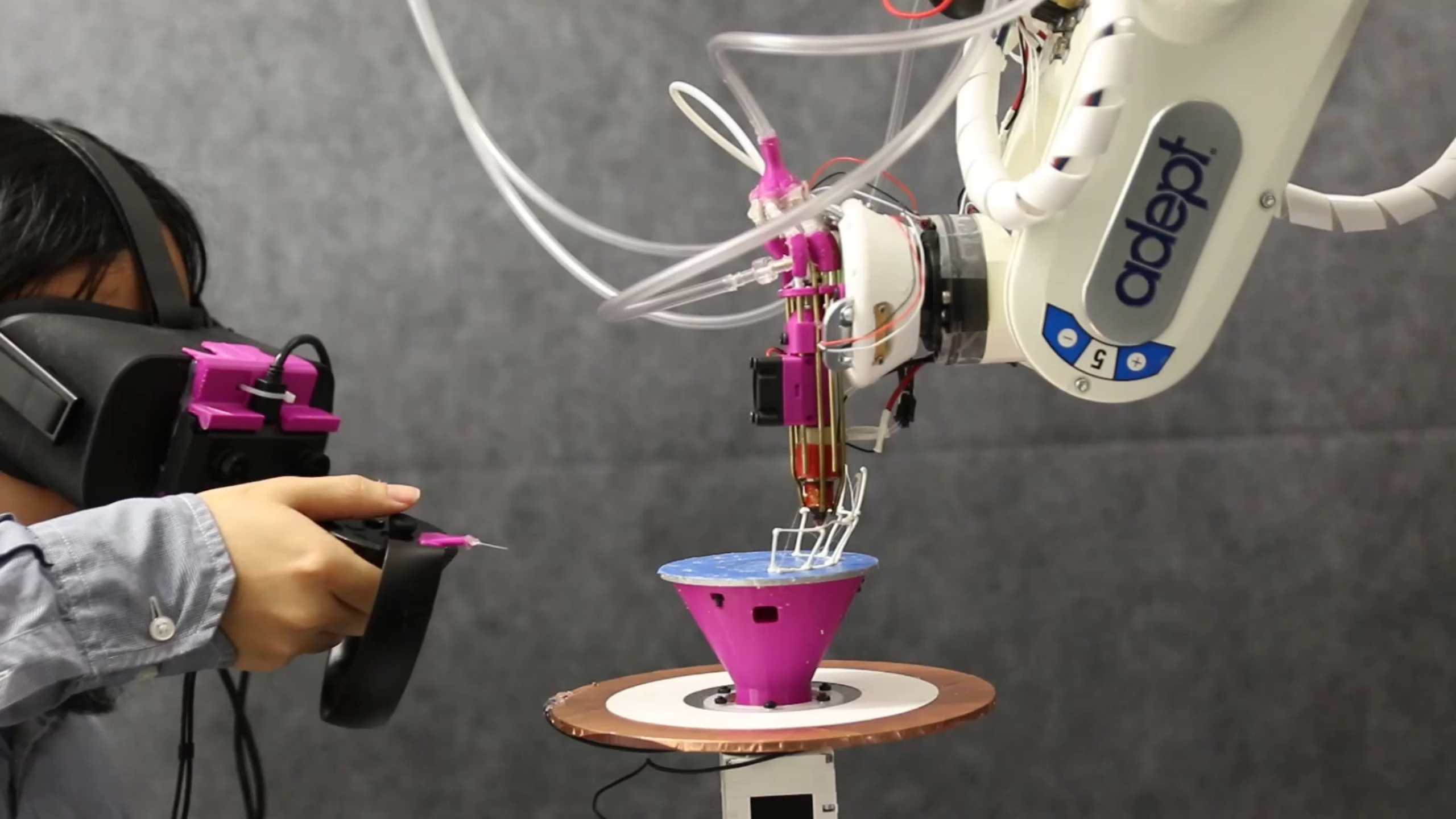


AR headset

Robotic 3D printer

AR controller

Rotating platform

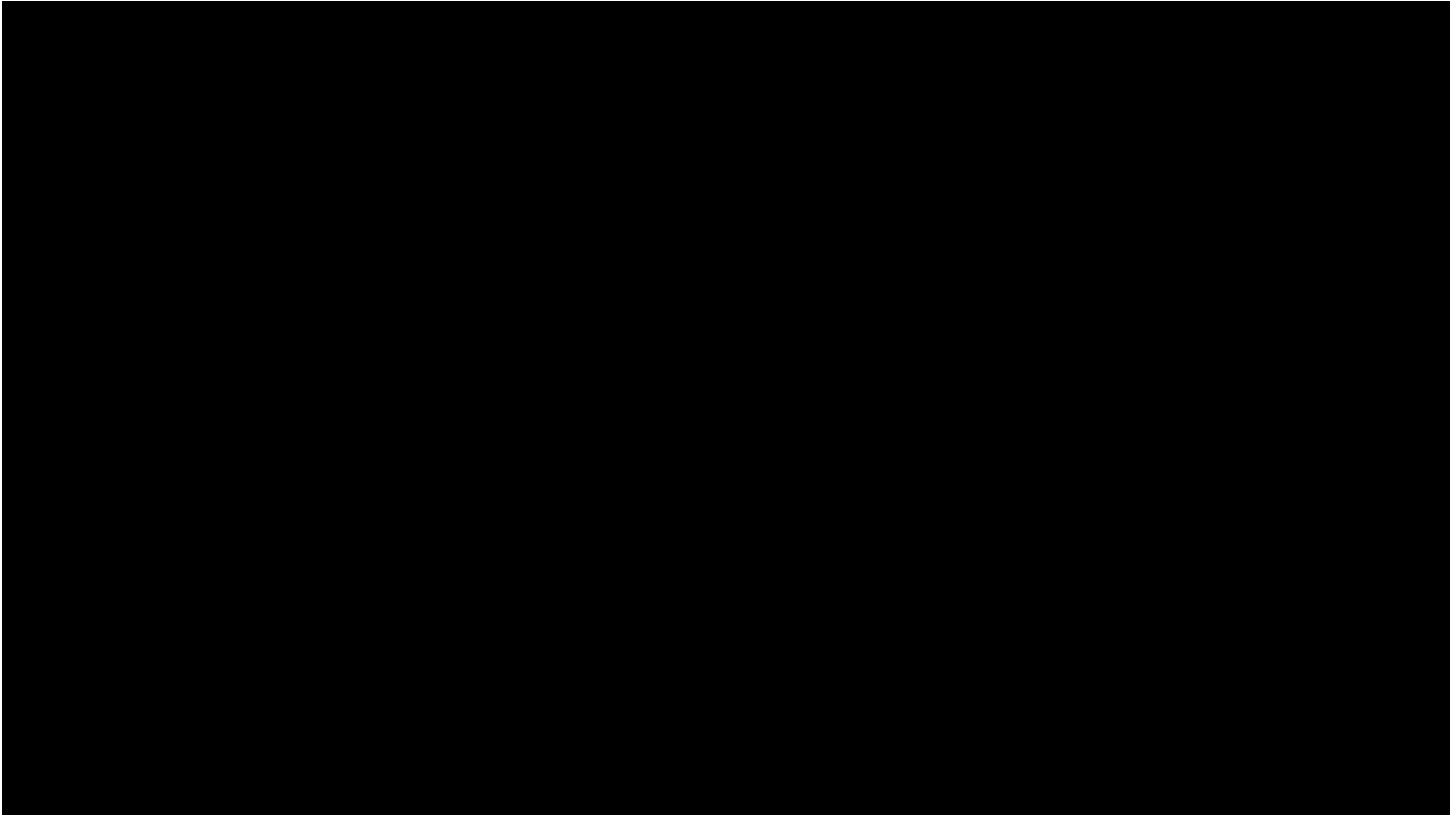


X3

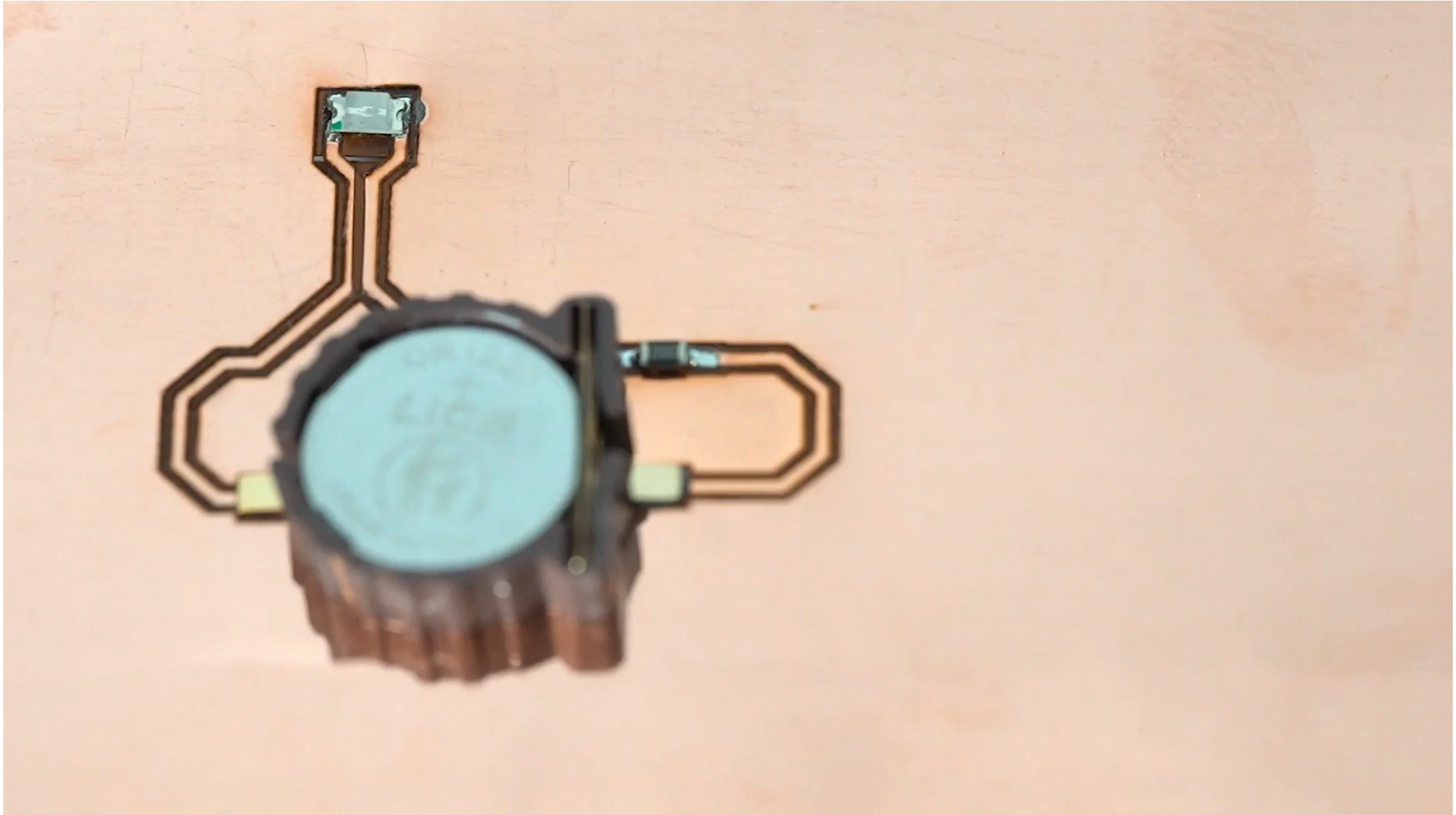


The designer creates a teapot body with *Revolve*

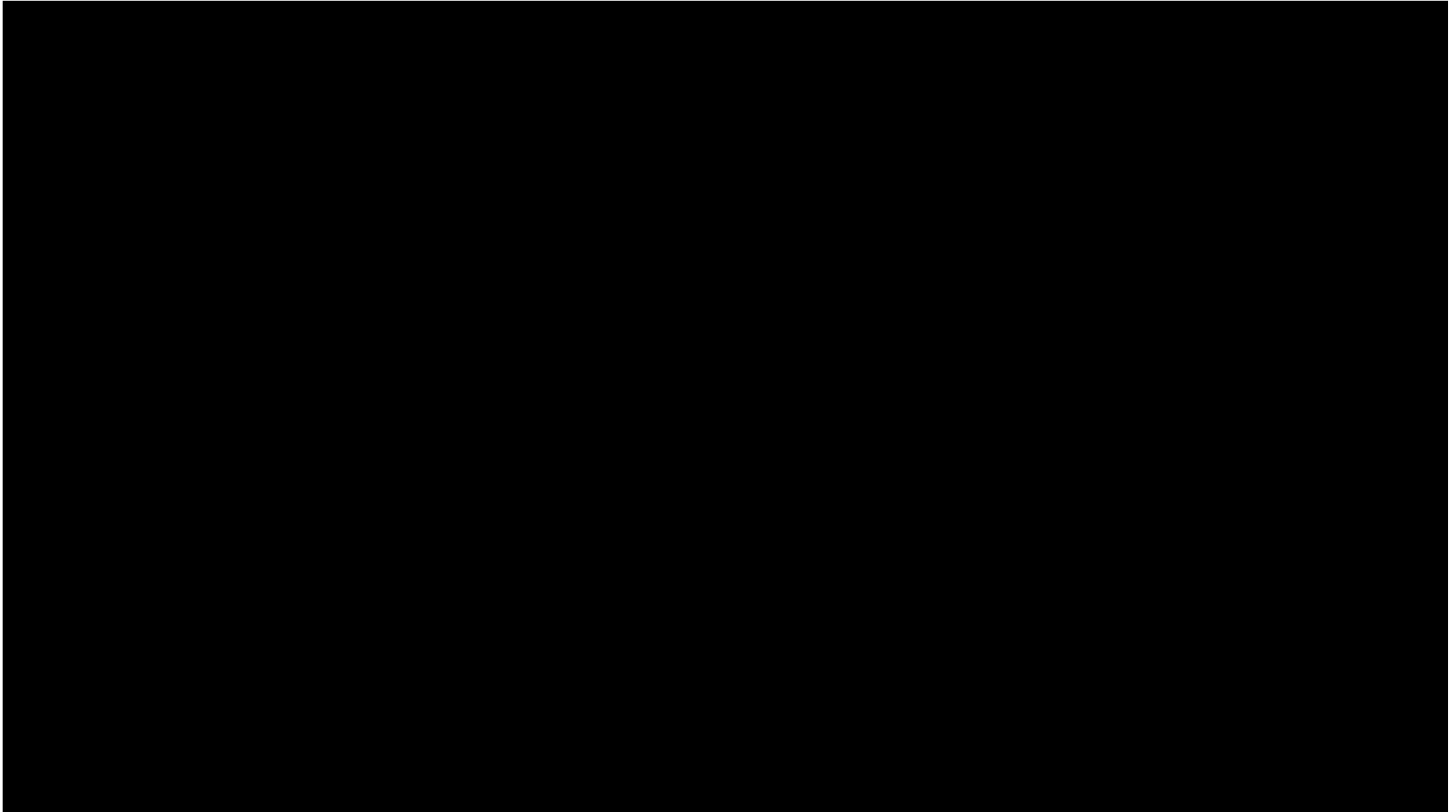
Adding haptics



Rapid prototyping of flexible circuit

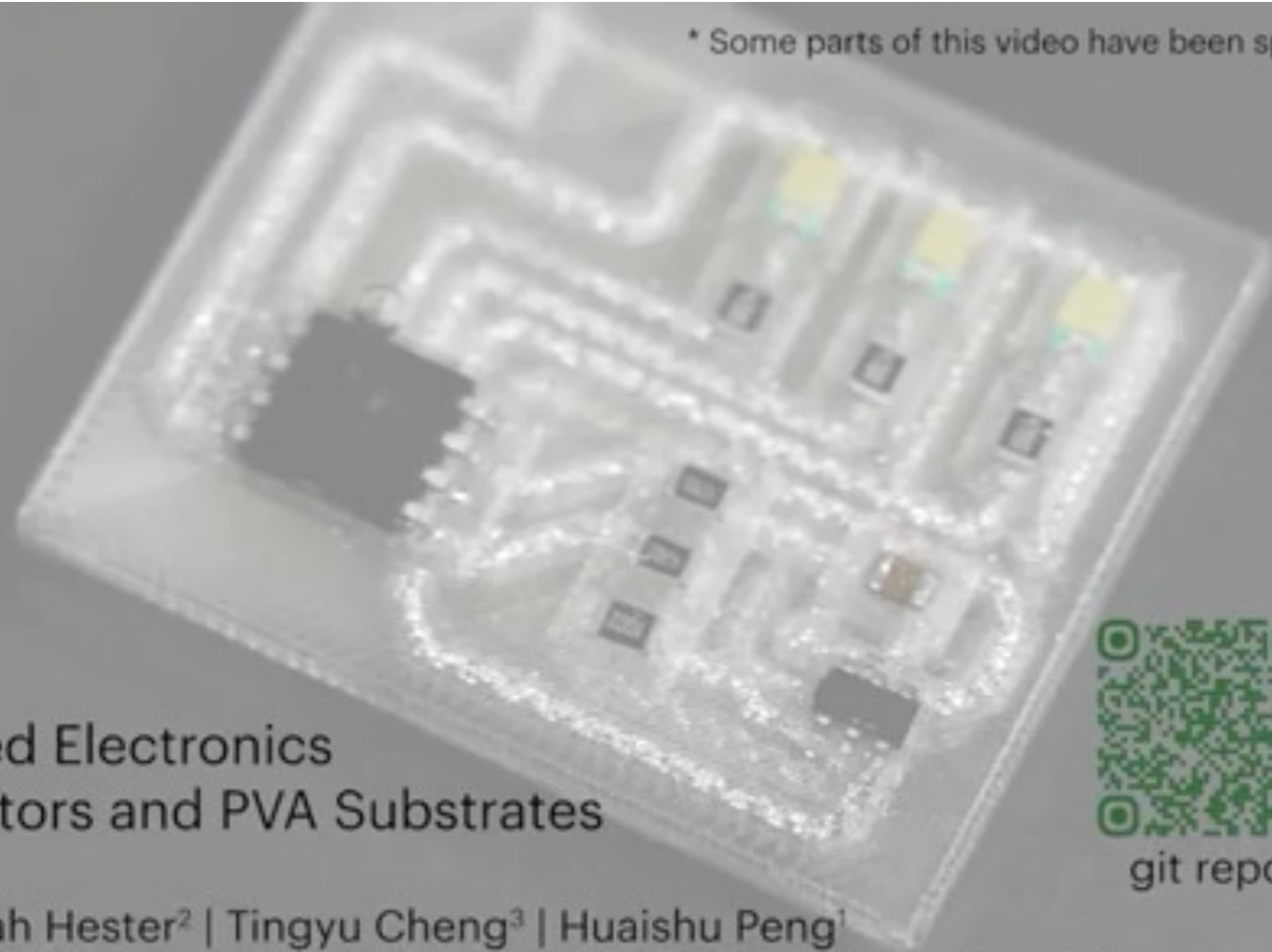


Rapid prototyping with reusing



Rapid prototyping with reusing

* Some parts of this video have been sped up.



DissolvPCB

Fully Recyclable 3D-Printed Electronics
with Liquid Metal Conductors and PVA Substrates



git repo

Zeyu Yan¹ | SuHwan Hong¹ | Josiah Hester² | Tingyu Cheng³ | Huaishu Peng¹

¹University of Maryland, College Park | ²Georgia Institute of Technology | ³University of Notre Dame



UNIVERSITY OF
MARYLAND

SMALL
ARTIFACTS
LAB



Georgia Institute
of Technology



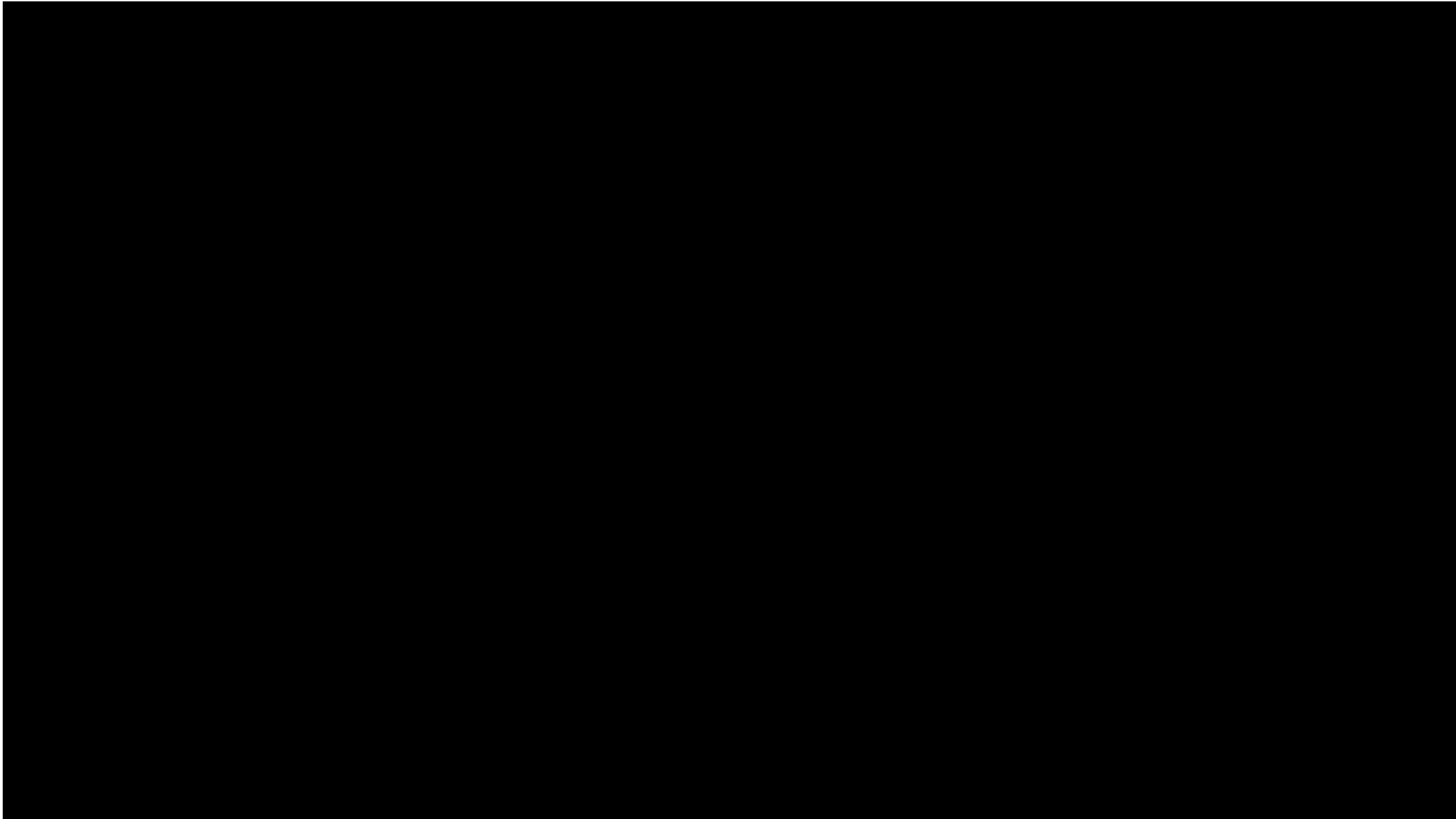
UNIVERSITY OF
NOTRE DAME

Beyond Fabrication?

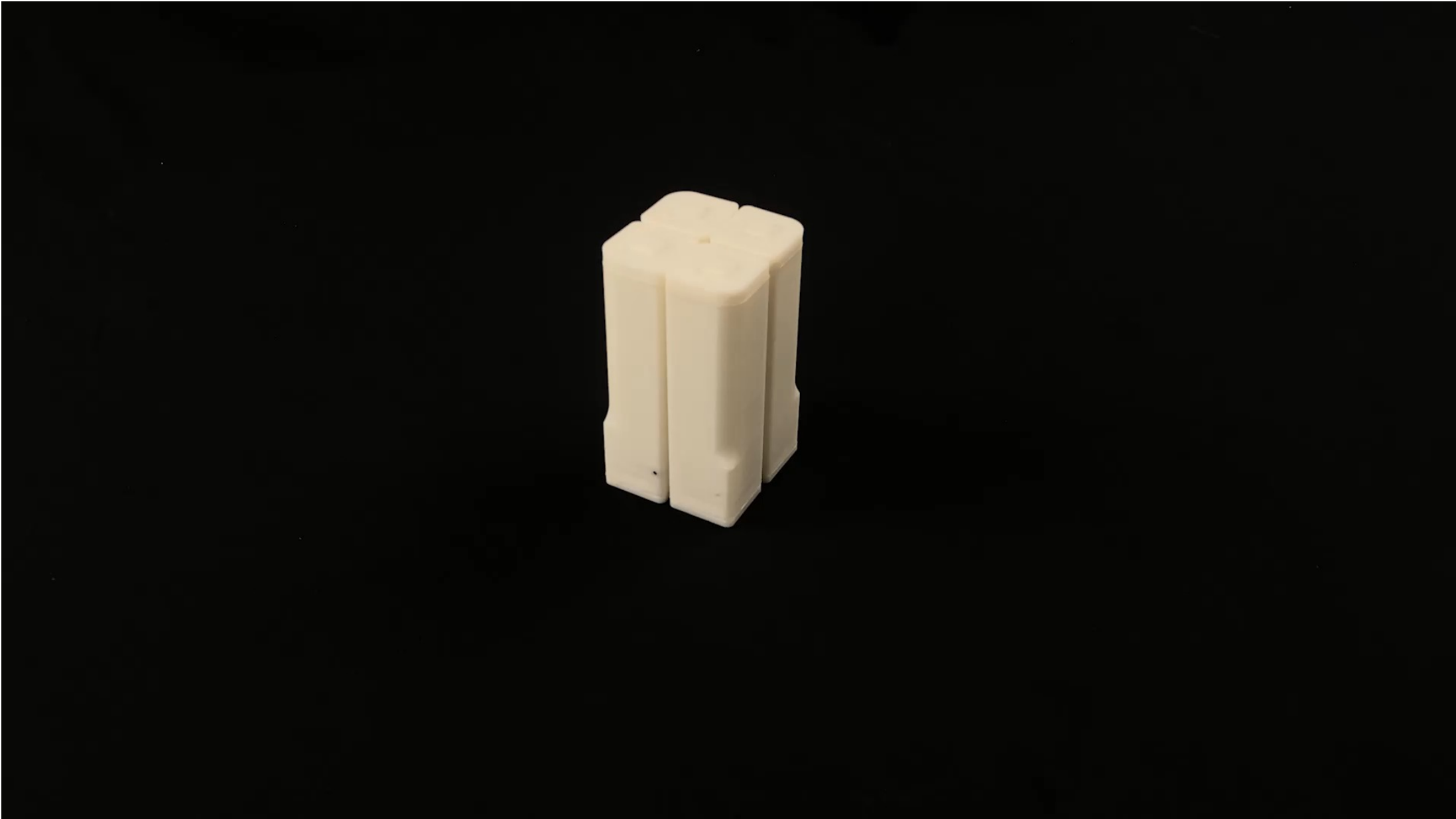


Accessibility

Support blind users to use public touchscreen kiosks



Support blind users to design websites



Lab space



IRB 0102

Now about this course

What this course is not about?

It's not about UX

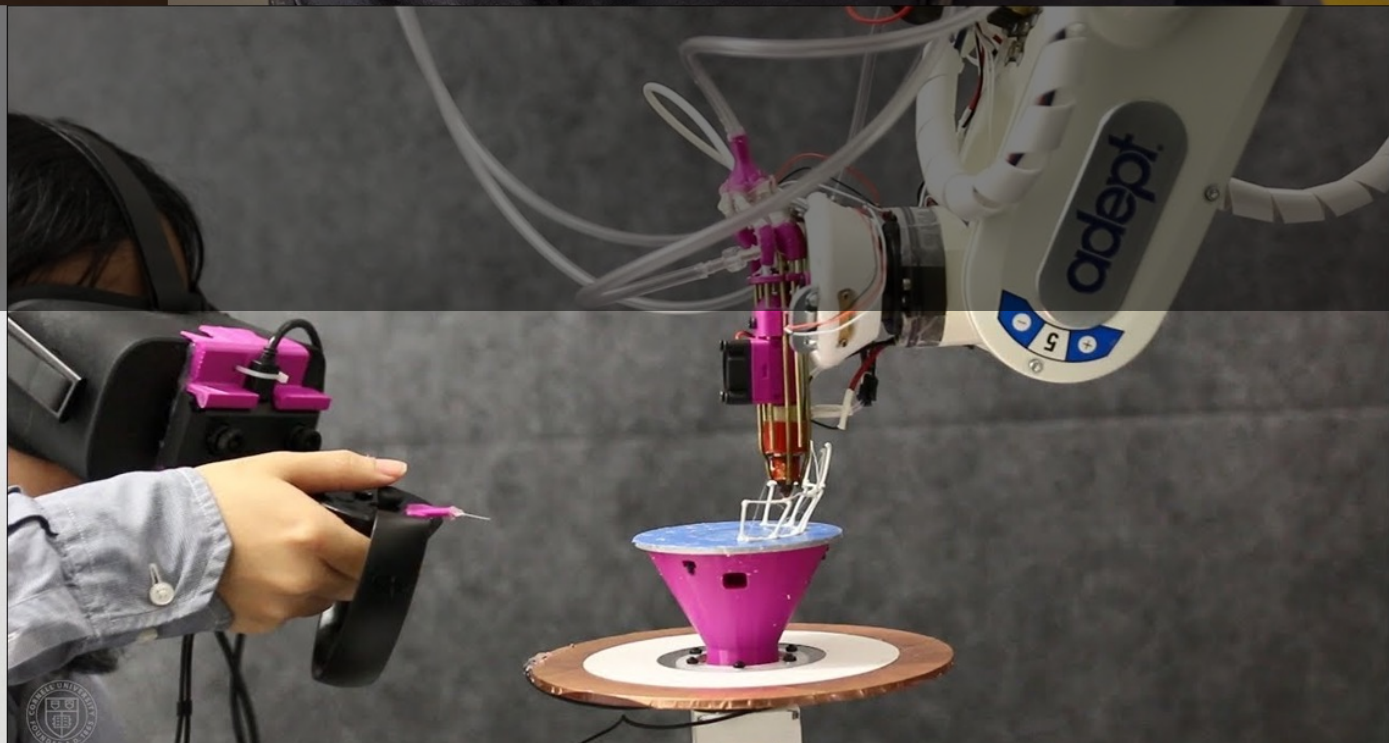
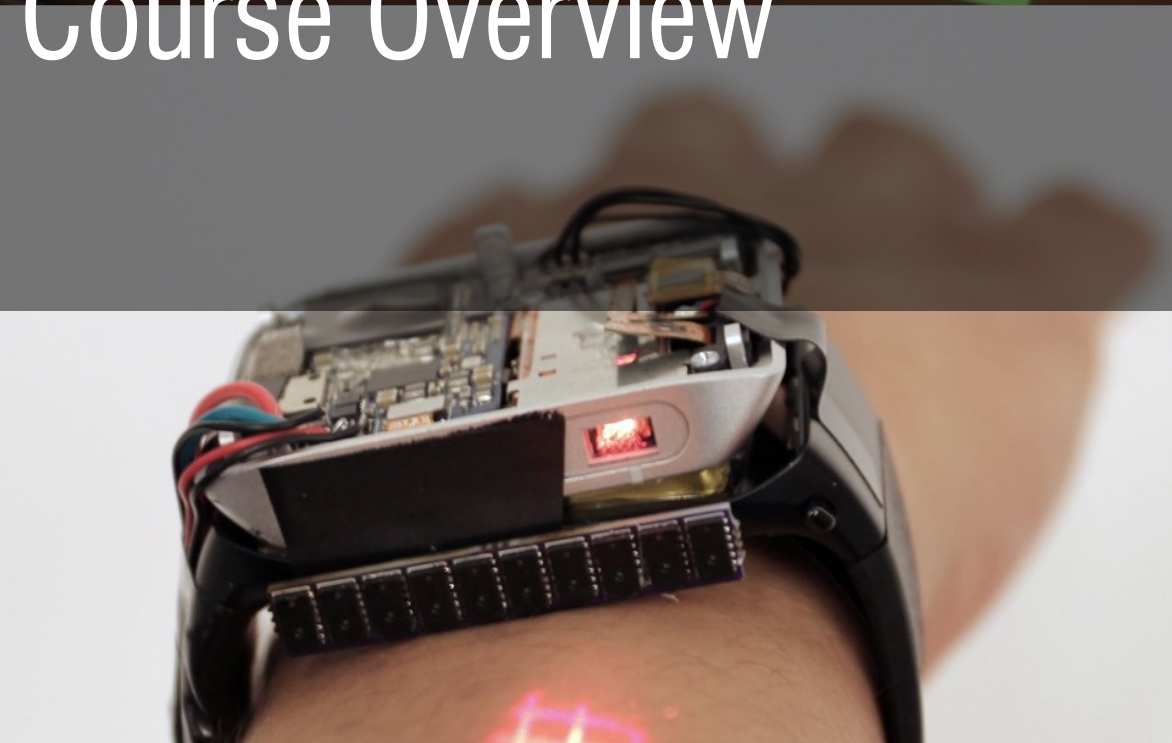
It's not about sketching

It's not about user research

It's not about user study

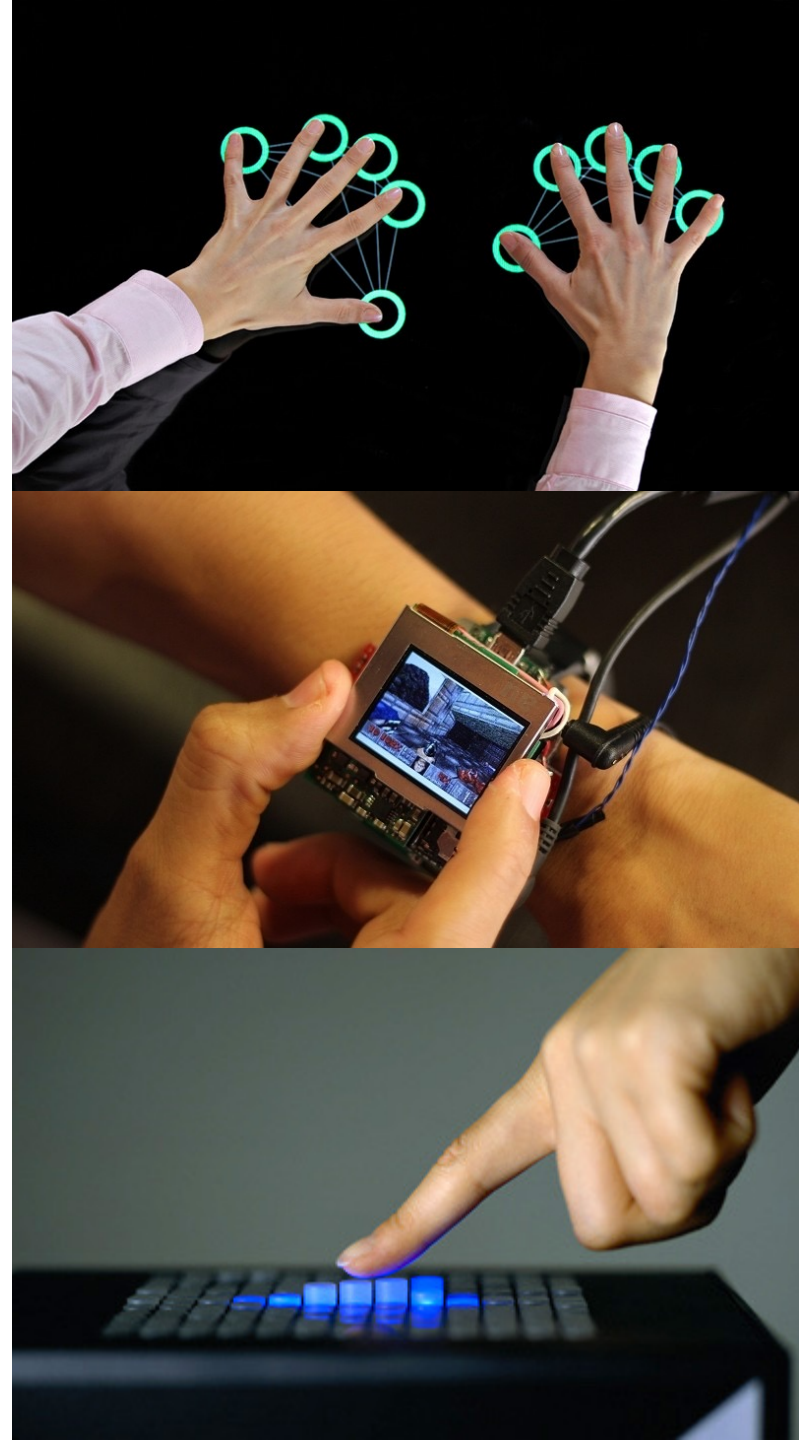


Course Overview



Learn

- Various interactive technologies
- Technologies behind the scene



Multi-touch Display

2007



“And we have **invented** a new technology called multi-touch, which is phenomenal.”
[0:33:33] - Steve Jobs

JEFFHAN

Low-Cost Multi-Touch Sensing through Frustrated Total Internal Reflection

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New York University
719 Broadway, New York, NY 10003
E-mail: jhan@mr1.nyu.edu



Figure 1: Simple examples of multi-touch interaction using our FTIR technique

ABSTRACT

This paper describes a simple, inexpensive, and scalable technique for enabling high-resolution multi-touch sensing on rear-projected interactive surfaces based on *frustrated total internal reflection*. We review previous applications of this phenomenon to sensing, provide implementation details, discuss results from our initial prototype, and outline future directions.

ACM Classification: H.5.2 [User Interfaces]: Input Devices and Strategies

General Terms: Human Factors

Keywords: multi-touch, touch, tactile, frustrated total internal reflection

INTRODUCTION

While touch sensing is commonplace for single points of contact, it is still difficult and/or expensive to construct a

We present a simple technique for robust multi-touch sensing at a minimum of engineering effort and expense. It is based on *frustrated total internal reflection (FTIR)*, a phenomenon familiar to both the biometric and robot sensing communities. It acquires true touch image information at high spatial and temporal resolutions, is scalable to large installations, and is well suited for use with rear-projection. It is not the aim of this paper to explore the multi-touch interaction techniques that this system enables, but rather to make the technology readily available to those who wish to do so.

RELATED WORK

A straightforward approach to multi-touch sensing is to simply utilize a plurality of discrete sensors, making an individual connection to each sensor as in the *Tactex MTC Express* [20]. They can also be arranged in a matrix configuration with some active element (e.g. diode, transistor) at each node, as in the device featured in Lee et al.'s seminal work [11], and also in Westerman and Elias's commercial *FingerWorks iGesturePad* [3][22].

UIST 2005

lasting impact award

Wearables

Best smartwatch 2020: the top wearables you can buy today

By [James Peckham](#) 7 hours ago

The absolute best smartwatches for your wrist



(Image credit: Samsung, Apple and Fossil)

Today's best smartwatch models can perform lots of tricks, like searching the internet with your voice, tracking your location with GPS or even monitoring your heart rate to protect your overall health. For some models, they can even do that without being paired to one of the [best phones](#) on the market.

We've tested the vast majority of the top-end wearables you can buy right now from the [Apple Watch](#) to Fitbits to [Garmin watches](#) to Tizen-sporting [Samsung watches](#). There's also [Wear OS](#) (you may have known that in its previous incarnation called Android Wear) which is Google's own wearable operating system in the vein of Apple's [watchOS](#) - you'll see it show up in a lot of these devices.

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2 **Switch Pro: developers tell us what they want from the rumored Switch upgrade**

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4 **PS5 release date, specs, news and rumors for Sony's PlayStation 5**

5 **How to watch the Star Wars movies in order**



APP NAVIGATION

SkinTrack: Using the Body as an Electrical Waveguide for Continuous Finger Tracking on the Skin

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¹Human-Computer Interaction Institute, ²Electrical and Computer Engineering Department
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ABSTRACT

SkinTrack is a wearable system that enables continuous touch tracking on the skin. It consists of a ring, which emits a continuous high frequency AC signal, and a sensing wristband with multiple electrodes. Due to the phase delay inherent in a high-frequency AC signal propagating through the body, a phase difference can be observed between pairs of electrodes. SkinTrack measures these phase differences to compute a 2D finger touch coordinate. Our approach can segment touch events at 99% accuracy, and resolve the 2D location of touches with a mean error of 7.6mm. As our approach is compact, non-invasive, low-cost and low-powered, we envision the technology being integrated into future smartwatches, supporting rich touch interactions beyond the confines of the small touchscreen.

Author Keywords

Finger tracking; waveguide; smartwatch; on-body interaction; around-device interaction; ADI

ACM Classification Keywords

H.5.2. [User interfaces] – Input devices and strategies.

INTRODUCTION

Small wearable devices—such as smartwatches and digital jewelry—are fast becoming viable computing platforms. However, their small size severely limits the user experience. For example, touchscreens on smartwatches suffer not only from a paucity of interactive surface area, but also must contend with significant finger occlusion. In general, the interfaces on these devices rely on basic input modalities (often four or fewer onscreen buttons, or even just directional swipes). In response, many research efforts have investigated how to leverage the area *around* devices to

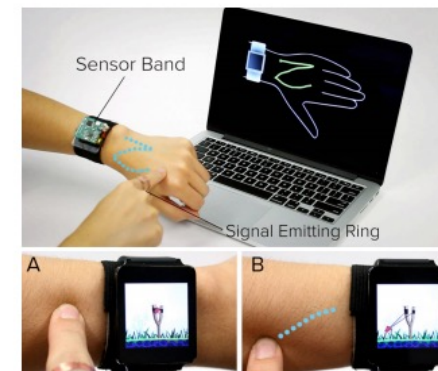


Figure 1. Our sensor band and signal-emitting ring allow the arm to be appropriated for continuous, on-skin touch tracking (top), expanding interaction beyond the small confines of a smartwatch touchscreen (bottom).

In this paper, we propose a novel sensing approach for appropriating the skin as an interactive, touch-tracking surface (Figure 1). Our system, *SkinTrack*, has two key components. First is a ring that emits an imperceptible and harmless 80MHz, 1.2Vpp AC signal into the finger on which it is worn. The second component is a wristband, worn on the opposite arm, and instrumented with a structured electrode pattern. When the user's finger touches the skin, the electrical signal propagates into the arm tissue and

CHI 2016

Zhang et.al. from CMU

XR



2012 Oculus Rift

XR



Haptic Turk: a Motion Platform Based on People

Lung-Pan Cheng, Patrick Lühne, Pedro Lopes, Christoph Sterz, and Patrick Baudisch
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ABSTRACT

Motion platforms are used to increase the realism of virtual interaction. Unfortunately, their size and weight is proportional to the size of what they actuate. We present *haptic turk*, a different approach to motion platforms that is light and mobile. The key idea is to replace motors and mechanical components with humans. All haptic turk setups consist of a *player* who is supported by one or more *human-actuators*. The player enjoys an interactive experience, such as a flight simulation. The motion in the player's experience is generated by the actuators who manually lift, tilt, and push the player's limbs or torso. To get the timing and force right, timed motion instructions in a format familiar from rhythm games are displayed on actuators' mobile devices, which they attach to the player's body. We demonstrate a range of installations based on mobile phones, projectors, and head-mounted displays. In our user study, participants rated not only the experience as player as enjoyable (6.1/7), but also the experience as an actuator (4.4/7). The approach of leveraging humans allows us to deploy our approach anytime anywhere, as we demonstrate by deploying at an art festival in the Nevada desert.

Author Keywords

Haptics; force-feedback; motion platform; immersion.

ACM Classification Keywords

H5.2 [Information interfaces and presentation]: User Interfaces. - Graphical user interfaces.

INTRODUCTION

For a long time, the key to immersion in interactive experience and games was sought in photorealistic graphics [8]. More recently, game makers made games more immersive by requiring players to physically enact the game such as with Wii (<http://wii.com>) and Kinect [26]. With graphics and user interaction now part of many games, many researchers argue that *haptics and motion* are the next step towards increasing immersion and realism in gaming

railing. Such events have been simulated using motion platforms [27]. Motion platforms are able to move one or more users around and have been used to add realism to flight simulators [22] and theme park rides.

Unfortunately, the size and weight of motion platforms tends to be proportional to what they actuate. As a result, motion platforms not only tend to be prohibitively expensive, but also large and heavy and thus stationary, limiting their use to arcades and lab environments.



Figure 1: Haptic turk allows producing motion experiences anywhere anytime. Here, the suspended player is enjoying an immersive hang gliding game. The four actuators create just the right physical motion to fill in the player's experience.

In this paper, we present *haptic turk*, a software platform that allows experiencing motion anywhere there is people. Its key idea is to substitute the motors and mechanical components of traditional motion platforms with humans.

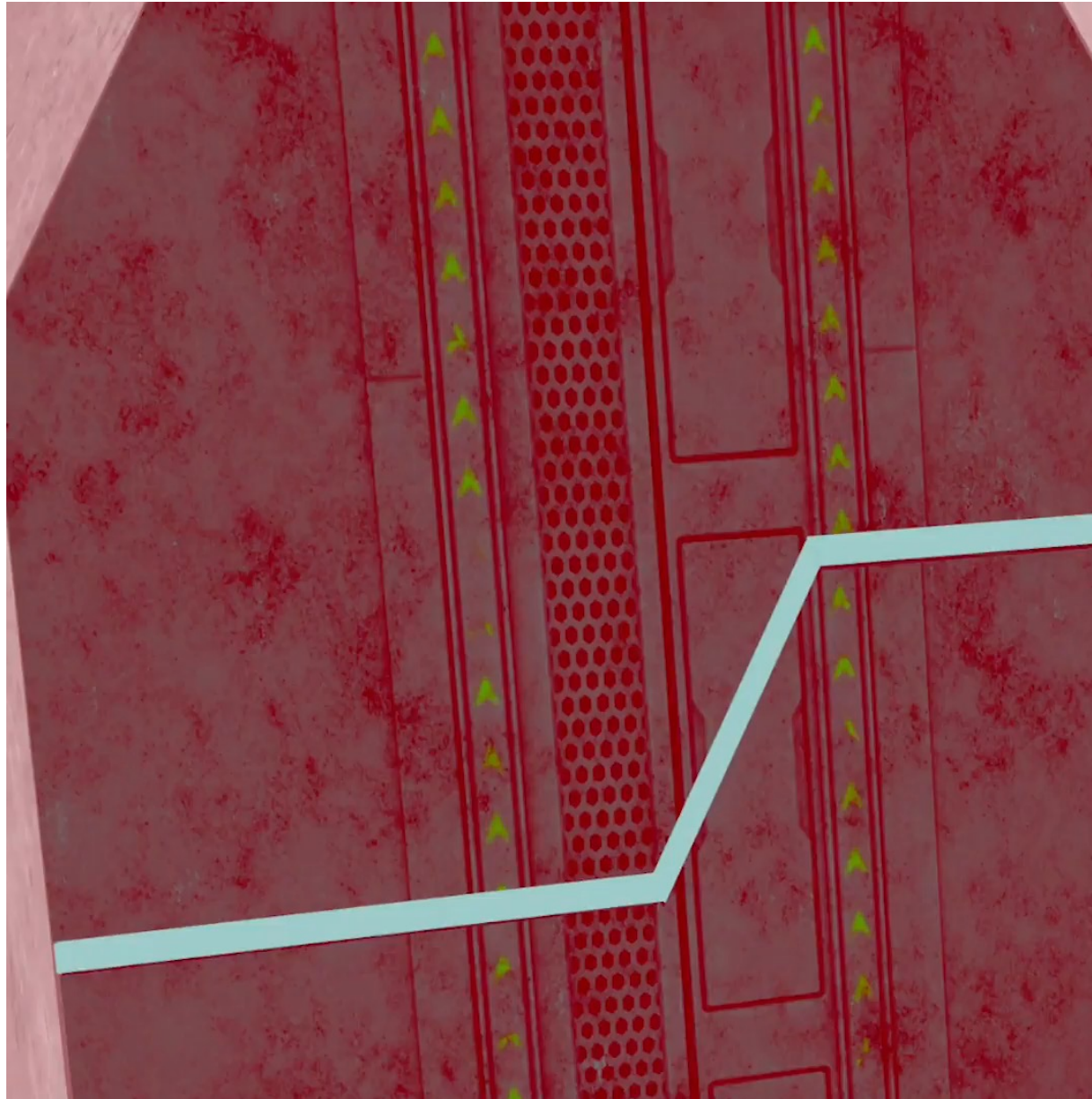
HAPTIC TURK

Haptic turk is a motion platform based on people. The name is inspired by the 18th century chess automation “The Turk” [20] that was powered by a human chess master.

CHI 2014

Cheng et.al. from HPI

XR



Chemical Haptics: Rendering Haptic Sensations via Topical Stimulants

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Figure 1: We propose a novel haptics approach, which we call *Chemical Haptics*, based on delivering topical stimulants to the user's skin. Upon absorbing these stimulants, receptors in the user's skin are chemically triggered, rendering distinct haptic sensations. To explore our approach in interactive contexts, such as VR, we engineered a self-contained wearable that delivers liquid stimulants. Here, it allows this VR user to feel four haptic sensations: (a) *Sanshool* creates tingling, which renders electric sparks emitted from a "short-circuiting" touchscreen on the arm; (b) *Lidocaine* creates numbing, which renders a malfunctioning arm interface by reducing tactile feedback as the user taps the buttons; (c) *Menthol* creates cooling, which renders cold winter air on the face; finally, (d) *Capsaicin* creates warming, which renders the hot air on the face from a nuclear reactor on the brink of meltdown.

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UIST '21, October 10–14, 2021, Virtual Event, USA
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ACM ISBN 978-1-4503-8635-7/21/10...\$15.00
<https://doi.org/10.1145/3472749.3474747>

ABSTRACT

We propose a new class of haptic devices that provide haptic sensations by delivering *liquid-stimulants* to the user's skin; we call this *chemical haptics*. Upon absorbing these stimulants, which contain safe and small doses of key active ingredients, receptors in the user's skin are chemically triggered, rendering distinct haptic sensations. We identified five chemicals that can render lasting haptic sensations: tingling (sanshool), numbing (lidocaine), stinging

UIST 2021

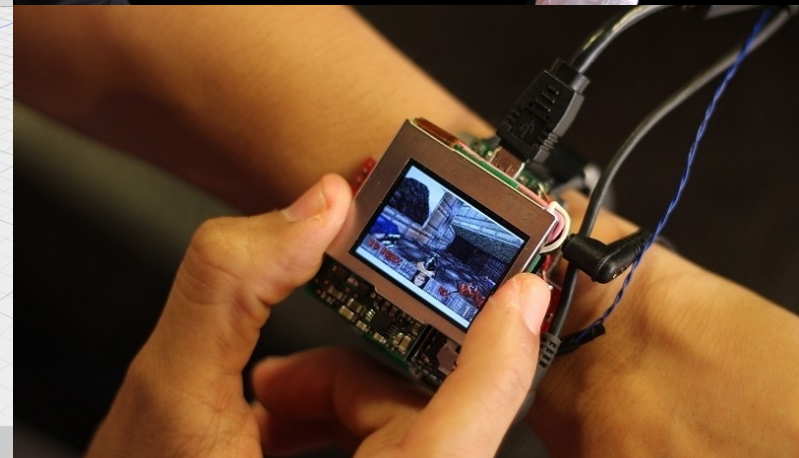
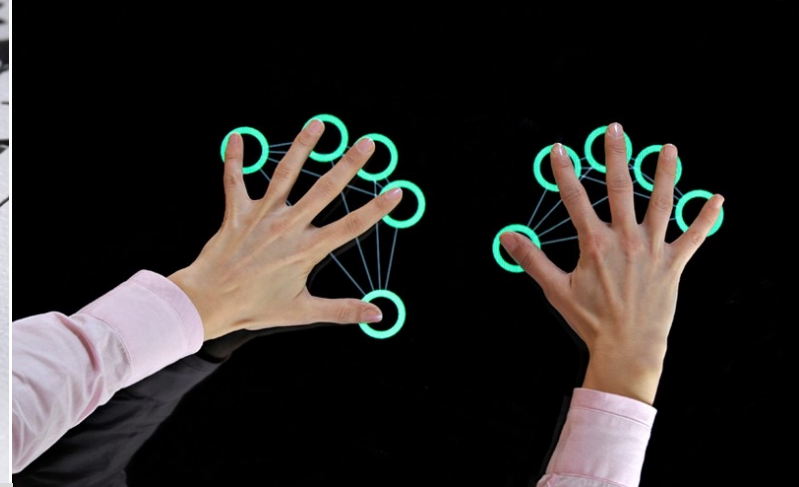
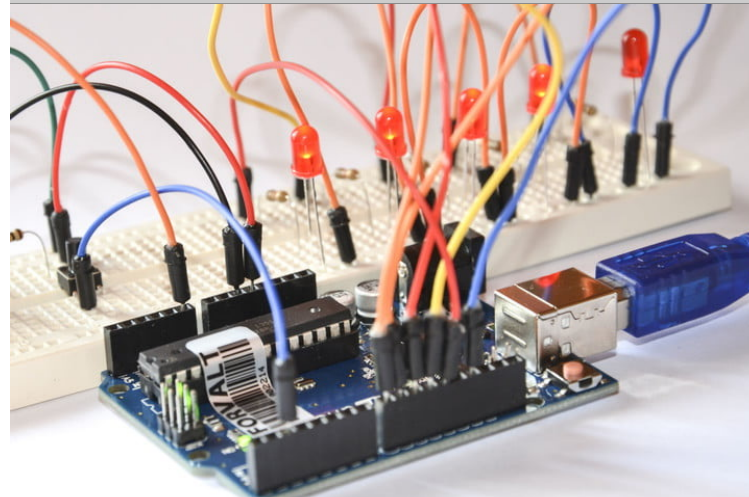
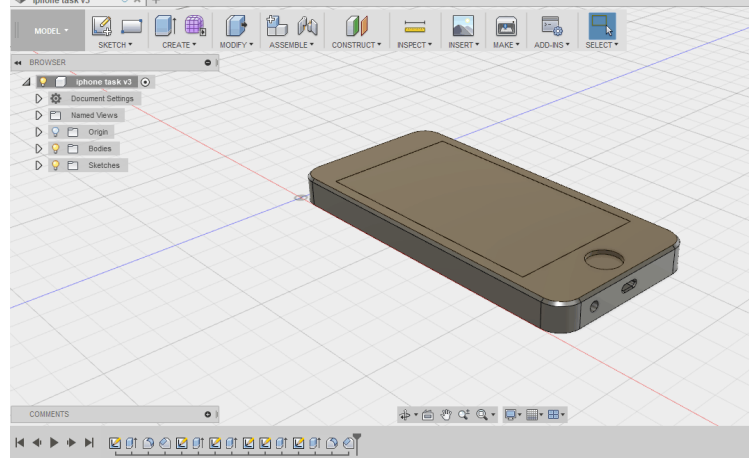
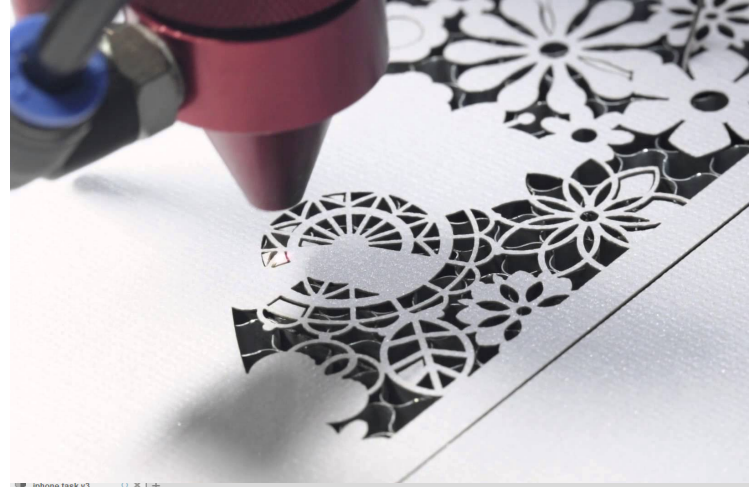
Lu et.al. from UChicago

Haptics
Fabrication
Tangible Interface

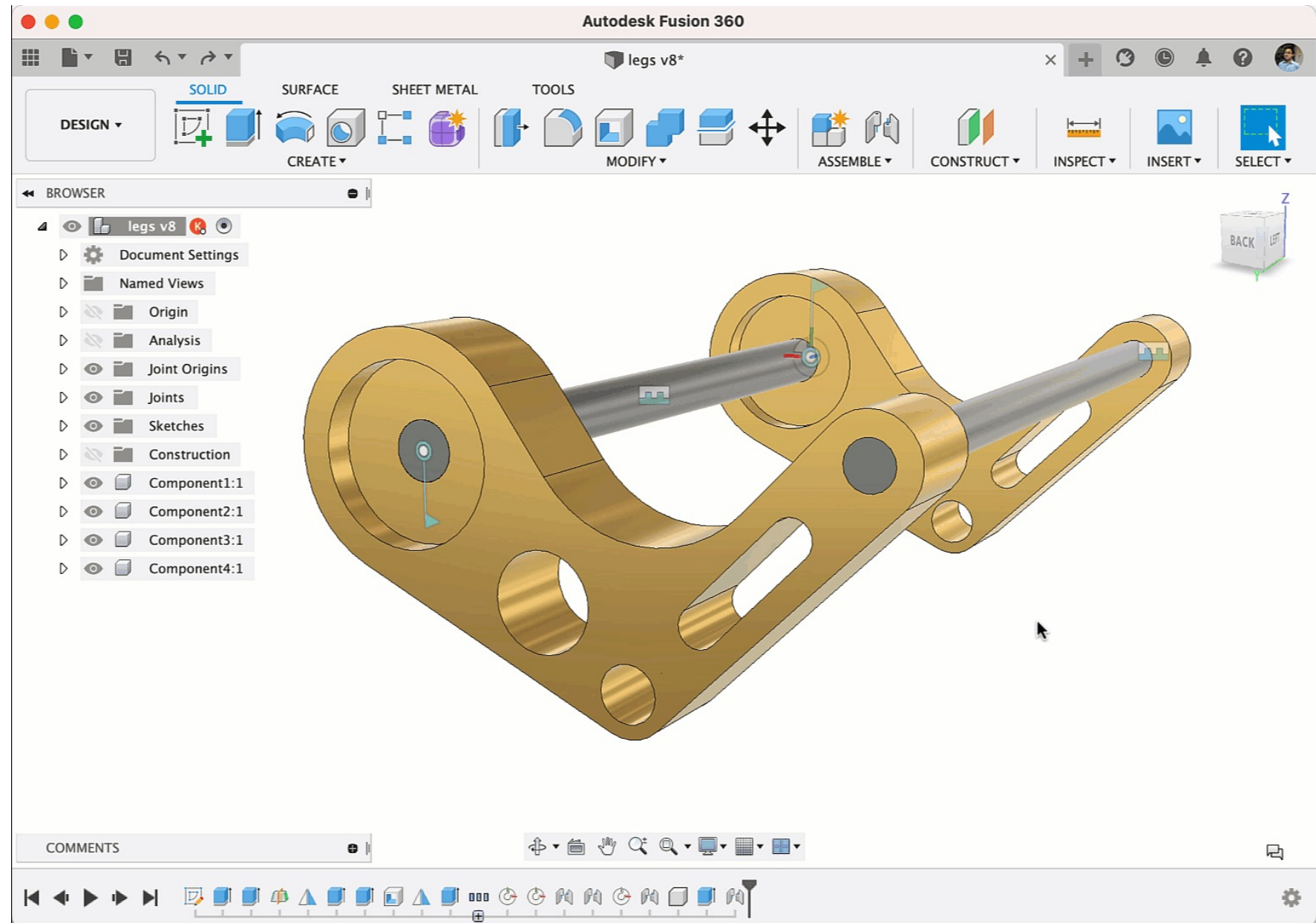
and more...
(well depends on the time we have)

Learn

- Various interactive technologies
- Technologies behind the scene
- Hands-on building skills
- Build interactive gadgets



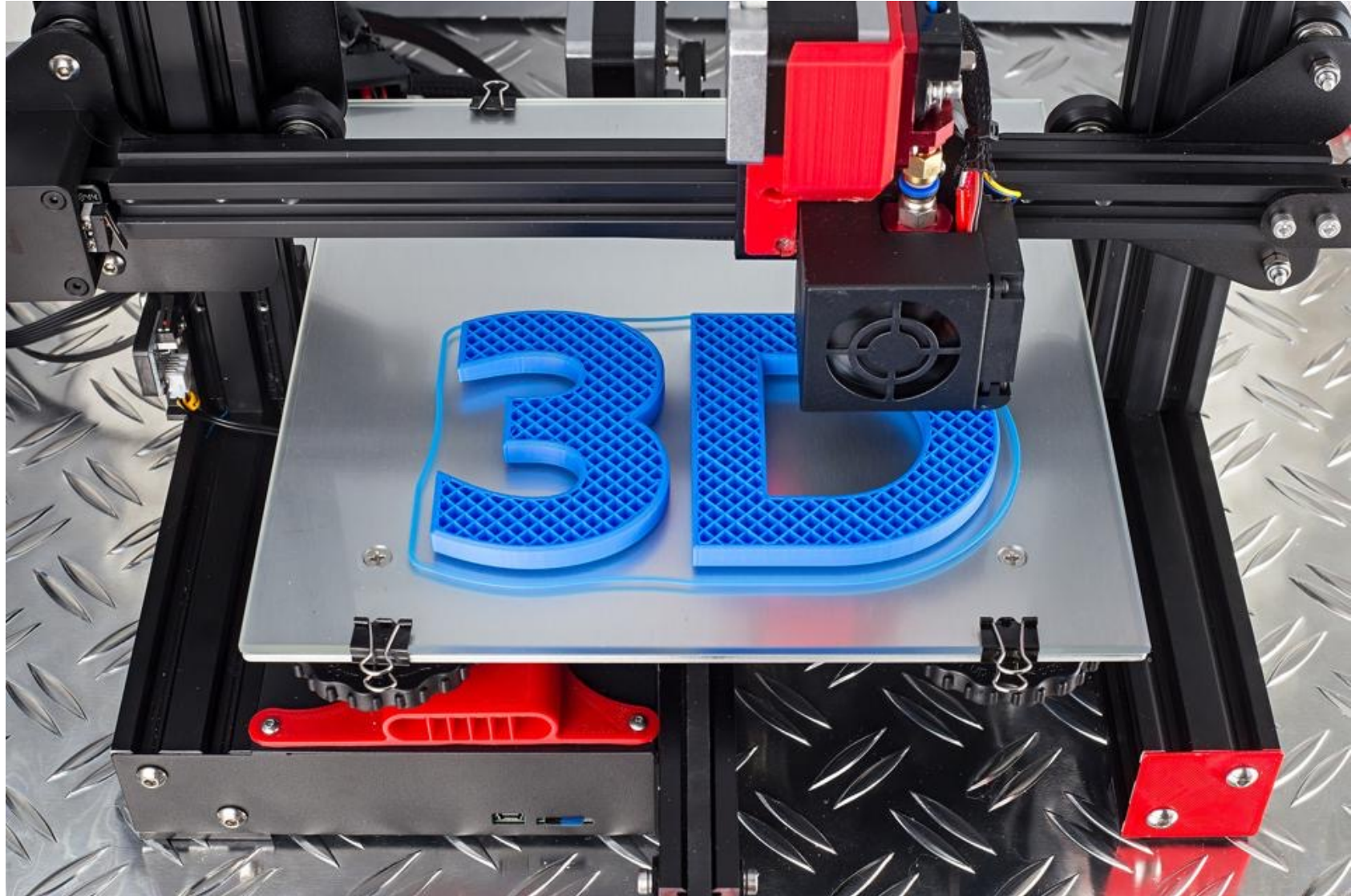
3D Modeling



Laser cutting

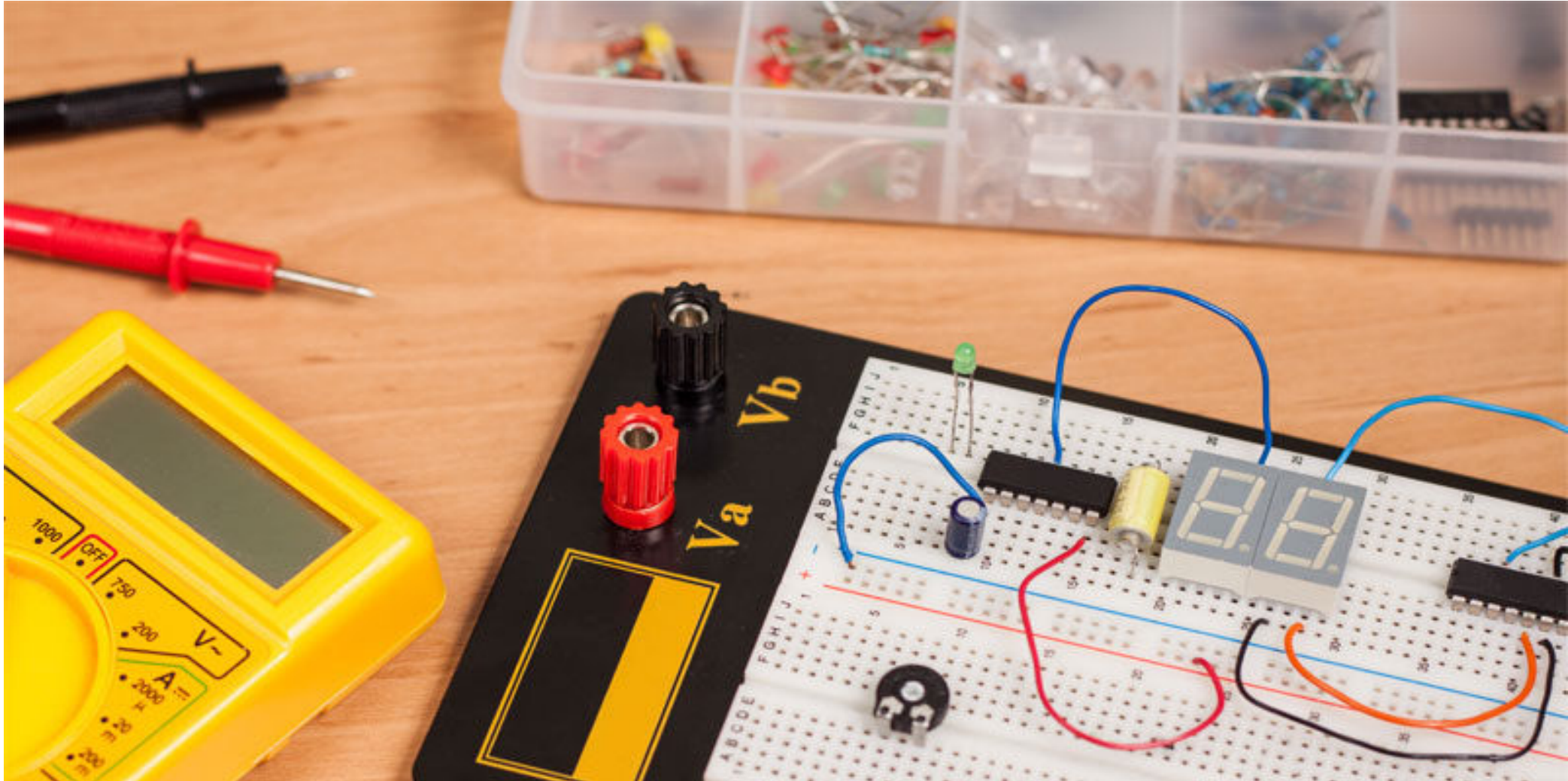


3D Printing



<https://www.forbes.com/sites/bernardmarr/2020/07/24/what-can-3d-printing-be-used-for-here-are-10-amazing-examples/?sh=2f19b9174d69>

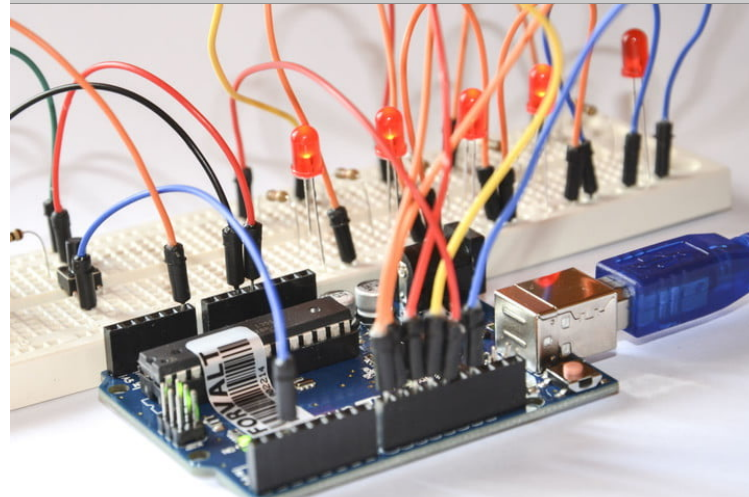
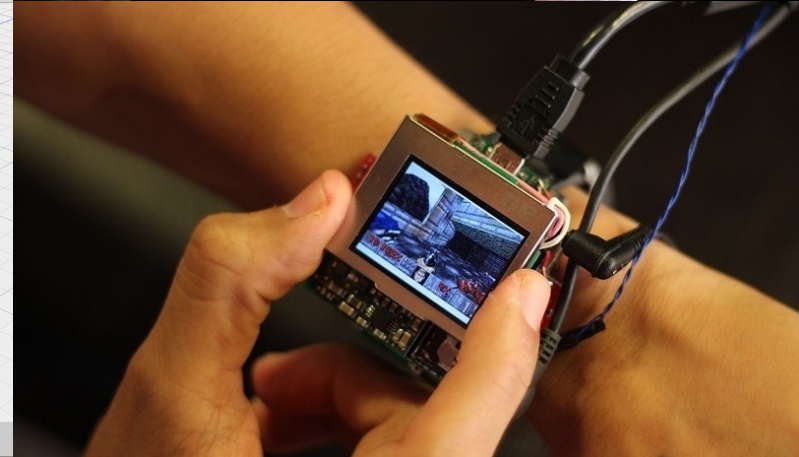
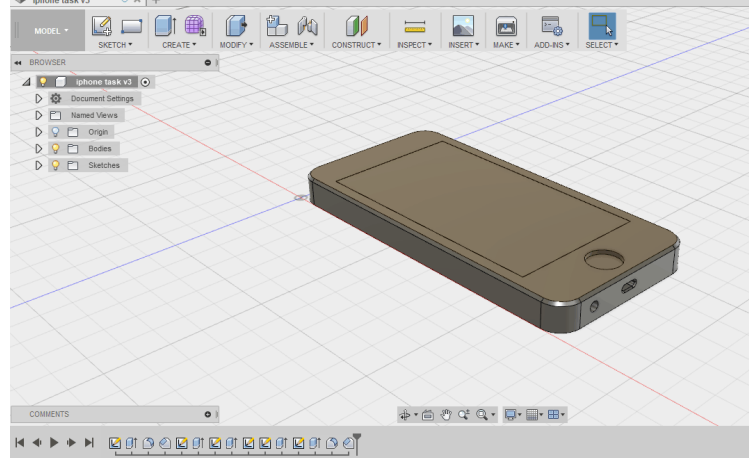
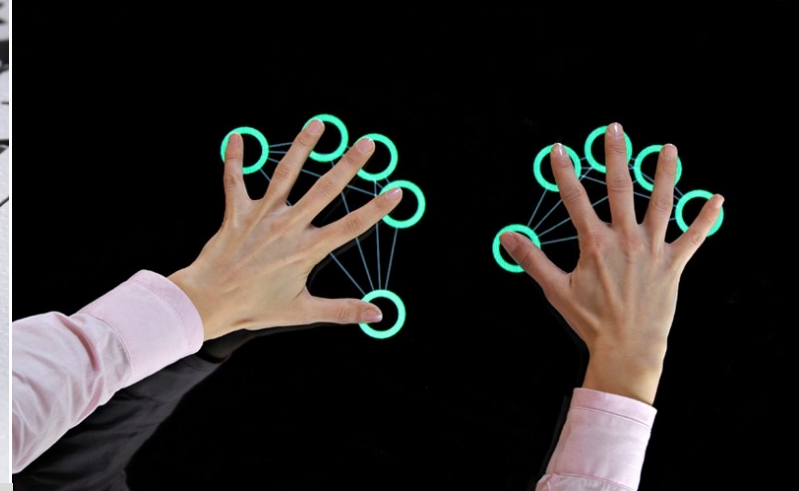
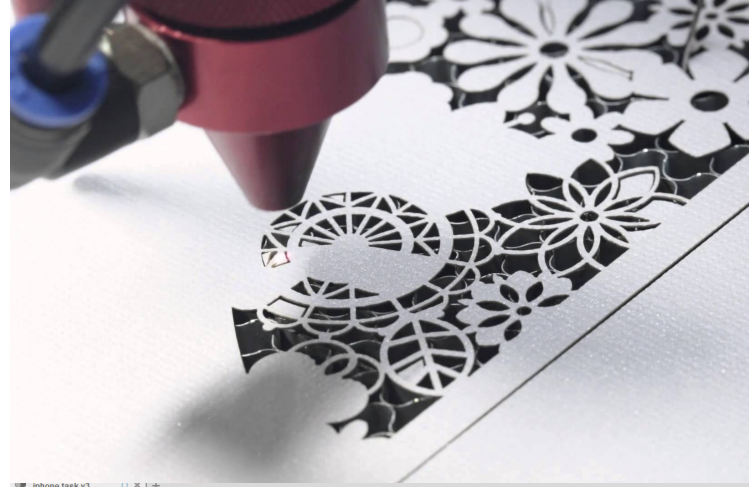
Electronics



<https://www.makerspaces.com/basic-electronics/>

Learn

- Various interactive technologies
- Technologies behind the scene
- Hands-on building skills
- Build interactive gadgets
- Mini robot competition
- Semester-long team projects



Mini robot competition

A team of 2 people will build a pipe-climbing robot in the span of 3 weeks, and take part in a competition to see whose robot (climb to the tallest point of a pipe/overcome all the obstacles/is the fastest)

More to come once we go through most of the hands-on building components.





Semester-long team project

You – a group of four - will build

1) an **interactive robot** or 2) a **tangible/haptic interface**
and present a **live** demo by the end of the semester

Semester-long team project

Requirement:

- a) Has to be a robot, a tangible interface, or a haptic device
- b) Involve both **hardware** + **software** components
- c) Need to be **interactive**
- d) Not strict replication – **novelty**

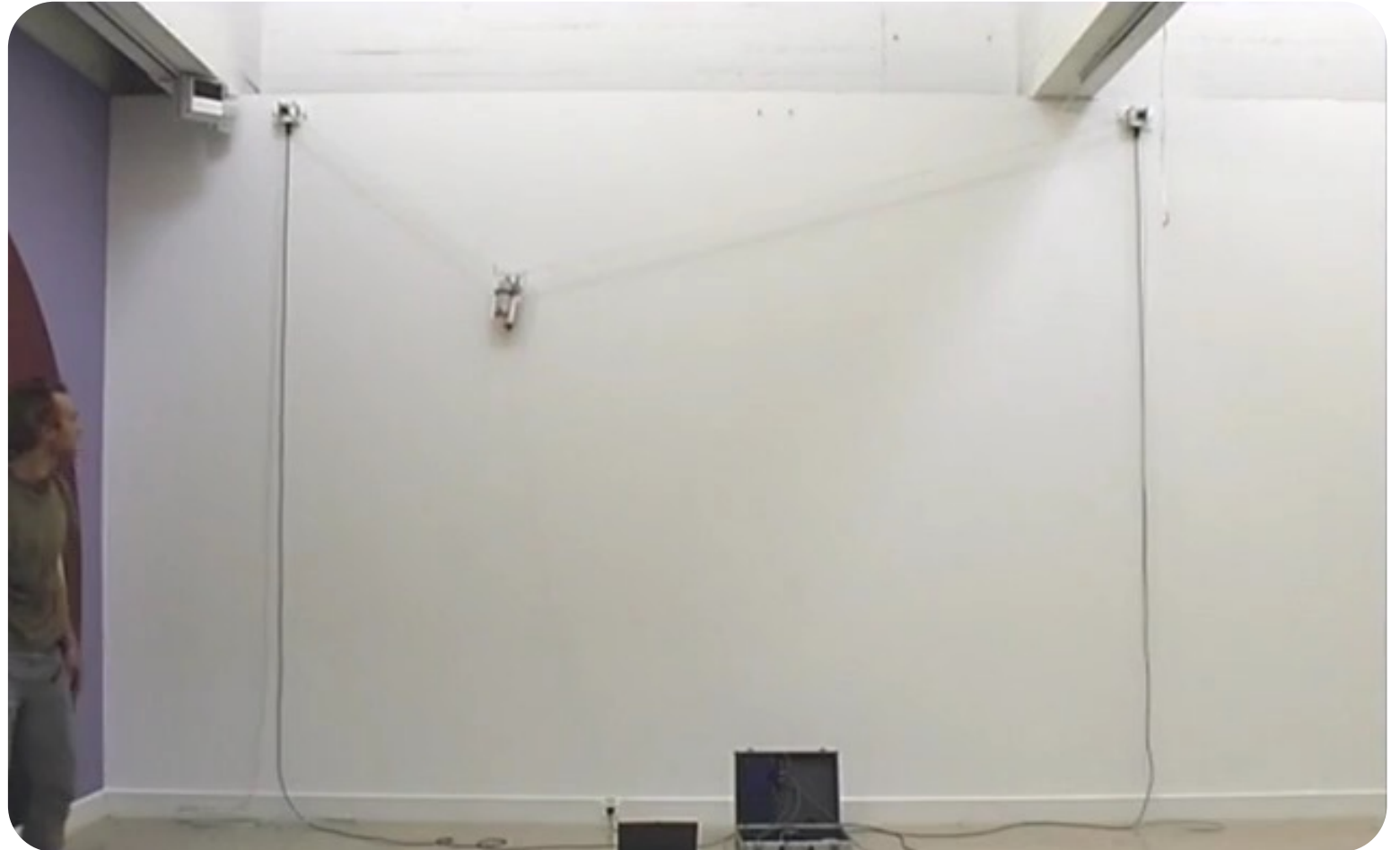
Semester-long team project

Deliveries:

- a) Final project report on **Hackster.IO Online documentation**
- b) Work-in-progress report, can be google doc
- c) **Videos** on YouTube
- d) Three milestone **presentations** (final presentation with **live** demo)

Possible topics

Robot drawing machine?



Jürg Lehni & Uli Franke 2002

Possible topics

Robot drawing machine?



Possible topics

Robot drawing machine?

Graffiti Fur: Turning Your Carpet into



Novel Hardware I

UIST'14, October 5–8, 2014, Honolulu, HI, USA

Graffiti Fur: Turning Your Carpet into a Computer Display

Yuta Sugiura¹, Koki Toda¹, Takayuki Hoshi²,
Youichi Kamiyama¹, Takeo Igarashi³ and Masahiko Inami¹

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² Center for Innovative Young Researchers, Nagoya Institute of Technology, Gokiso, Showa, Nagoya, 466-8555 Japan
³ Department of Computer Science, The University of Tokyo, 7-3-1 Hongo, Bunkyo, Tokyo, 113-0033 Japan
{y-sugiura | kmd-std.8979 | kamiyama | inami}@kmd.keio.ac.jp,
star@nitech.ac.jp, takeo@acm.org

ABSTRACT

We devised a display technology that utilizes the phenomenon whereby the shading properties of fur change as the fibers are raised or flattened. One can erase drawings by first flattening the fibers by sweeping the surface by hand in the fiber's growth direction, and then draw lines by raising the fibers by moving the finger in the opposite direction. These material properties can be found in various items such as carpets in our living environments. We have developed three different devices to draw patterns on a “fur display” utilizing this phenomenon: a roller device, a pen device and pressure projection device. Our technology can turn ordinary objects in our environment into rewritable displays without requiring or creating any non-reversible modifications to them. In addition, it can be used to present large-scale image without glare, and the images it creates require no running costs to maintain.

Author Keywords

Fur Display; BRDF; Living Environment;

ACM Classification Keywords

H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

INTRODUCTION

Computer displays play an important role in connecting the information world and the real world. In the era of ubiquitous computing, it is essential to be able to access information in a fluid way and non-obstructive integration of displays into our living environment is a basic requirement to achieve it. However, common displays such as LCDs are not ideal for continuous use in living environments; they occupy considerable space, emit glaring



Figure 1: The devices convert your carpet into a computer display.

light that disturbs human vision and consume electric power.

On the other hand, projection-type displays are suitable for ubiquitous environments because they can project images onto any surface in the living environment [4, 12, 24]. Accordingly, the environment does not require large-scale modifications. In addition, it is possible to switch the projection rapidly and project colorful images. However, projectors have disadvantages. For example, the projected images cause glare and are hard to see in a bright room. In addition, the electricity costs of continuously projecting images are quite high.

There have been many proposals and implementations of non-emissive displays that can easily be integrated into living environments without causing any glare, such as E-ink¹ and wooden displays [18]. They do not require

UIST 2014

Sugiura et.al.

Possible topics

XR + robot



Flying simulator with haptic feedback. Cade Haley, Madison Razook and Kyle Gronberg
Course Project@CU Boulder with Prof. Daniel Leithinger, 2018

Possible topics

XR + robot

CHI 2020 Paper

CHI 2020, April 25–30, 2020, Honolulu, HI, USA

FaceHaptics: Robot Arm based Versatile Facial Haptics for Immersive Environments

Alexander Wilberz
Hochschule Bonn-Rhein-Sieg
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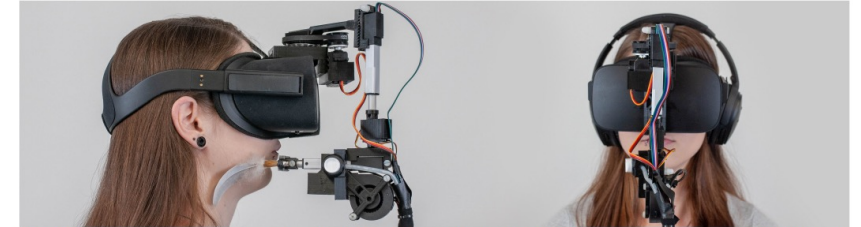


Figure 1. The FaceHaptics system, showing a side and frontal view of the setup for face haptic feedback, affording various sensations including touch, texture, warmth, air flow, or wetness. The left image depicts one of many possible touch/texture feedback elements, which can easily be exchanged.

ABSTRACT

This paper introduces FaceHaptics, a novel haptic display based on a robot arm attached to a head-mounted virtual reality display. It provides localized, multi-directional and movable haptic cues in the form of wind, warmth, moving and single-point touch events and water spray to dedicated parts of the face not covered by the head-mounted display. The easily extensible system, however, can principally mount any type of compact haptic actuator or object. User study 1 showed that users appreciate the directional resolution of cues, and can judge wind direction well, especially when they move their head and wind direction is adjusted dynamically to compensate for head rotations. Study 2 showed that adding FaceHaptics cues to a VR walkthrough can significantly improve user experience, presence, and emotional responses.

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Author Keywords

Haptics; robot arm; immersive environments; virtual reality; user study; perception; presence; emotion

CCS Concepts

•Human-centered computing → Human computer interaction (HCI); Haptic devices; User studies;

INTRODUCTION AND MOTIVATION

Over the last decade, Virtual Reality (VR) systems have been massively improved, in particular driven by the gaming industry but increasingly also other industry sectors. Predominantly, advances have been made in providing affordable yet high quality visual displays. However, non-visual cues can be a key factor in immersive systems, for example to improve overall simulation and perceptual fidelity [36] or to invoke emotional reactions [14, 27]. While rendering audio cues is well supported, haptic feedback is still challenging, and foremost repeated towards the hands [21]. In this paper, we look at how

CHI 2020

Wilberz et.al.

Possible topics

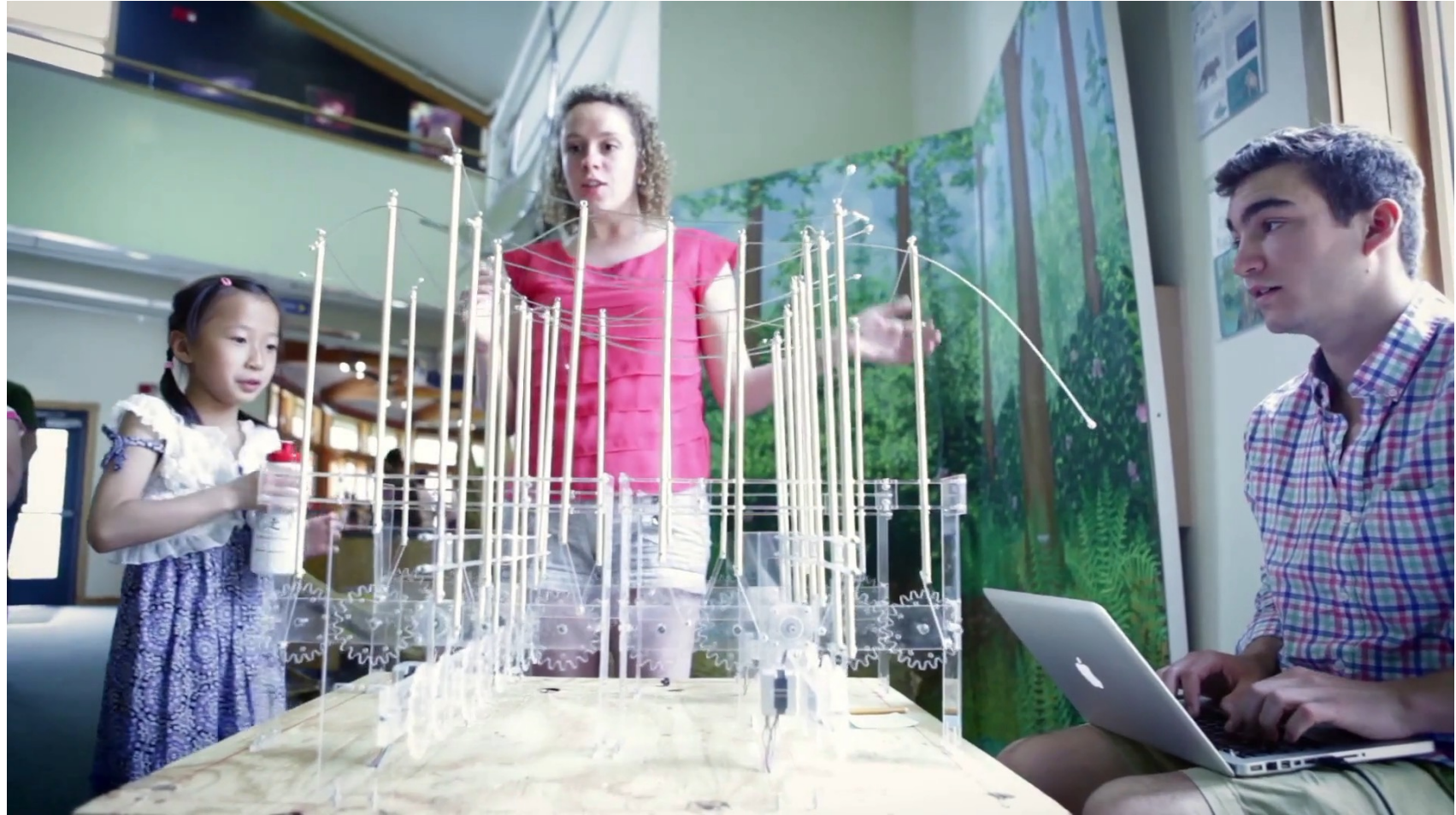
Kinetic Display?



Kinetic Sculpture BMW. JOACHIM SAUTER
2008

Possible topics

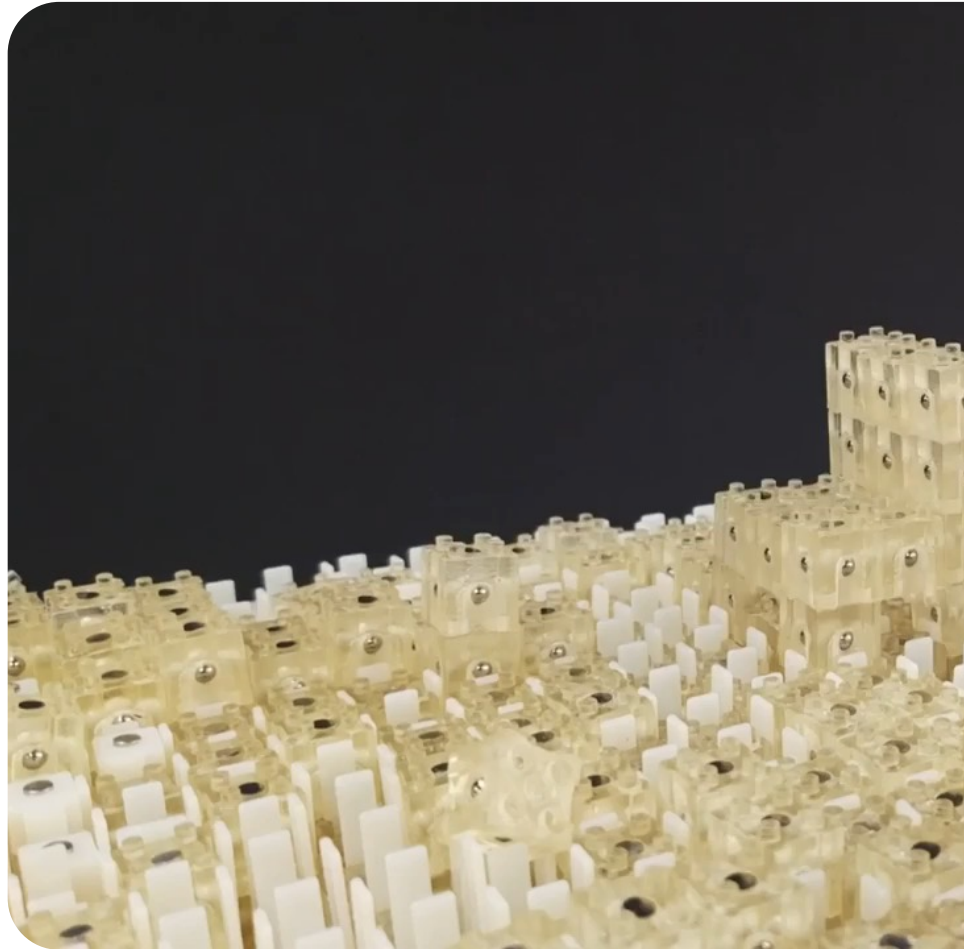
Kinetic Display?



Kinetic Sculpture. Megan Baker, Andrea Cameron and Aleksey Polesskiy.
Cornell INFO4320 Course Project 2014

Possible topics

Kinetic Display?



Dynablock: Dynamic 3D Printing for Instant and Reconstructable Shape Formation

Ryo Suzuki¹, Junichi Yamaoka², Daniel Leithinger¹, Tom Yeh¹
Mark D. Gross¹, Yoshihiro Kawahara², Yasuaki Kakehi²

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Figure 1. Dynablock is a rapid and reconstructable shape formation system, comprised of a large number of small physical elements. A) Dynablock's shape consists of 9 mm blocks which can be connected with omni-directional magnets. B-D) Dynablock leverages the 24 x 16 pin-based shape display as a parallel assembler of blocks, Dynablock is able to construct three-dimensional shapes in seconds. E) The example shows the output of a miniature model of table and a chair. The constructed shape is graspable and reconstructable.

ABSTRACT

This paper introduces Dynamic 3D Printing, a fast and reconstructable shape formation system. Dynamic 3D Printing assembles an arbitrary three-dimensional shape from a large number of small physical elements. It can also disassemble the shape back to elements and reconstruct a new shape. Dynamic 3D Printing combines the capabilities of 3D printers and shape displays: Like conventional 3D printing, it can generate arbitrary and graspable three-dimensional shapes, while allowing shapes to be rapidly formed and reformed as in a shape display. To demonstrate the idea, we describe the design and implementation of Dynablock, a working prototype of a dynamic 3D printer. Dynablock can form a three-dimensional shape in seconds by assembling 3,000 9 mm blocks, leveraging a 24 x 16 pin-based shape display as a parallel assembler. Dynamic 3D printing is a step toward achieving our long term vision in which 3D printing becomes an interactive medium, rather than the means for fabrication that it is today. In this paper we explore possibilities for this vision by illustrating application scenarios that are difficult to achieve with conventional 3D printing or shape display systems.

CCS Concepts

•Human-centered computing → Interaction devices;

INTRODUCTION

What if 3D printers could form a physical object in seconds? What if the object, once it is no longer needed, could quickly and easily be disassembled and reconstructed as a new object? Today's 3D printers take hours to print objects, and output a single static object. However, we envision a future in which 3D printing could instantly create objects from reusable and reconstructable materials.

With these capabilities, a 3D printer would become an interactive medium, rather than merely a fabrication device. For example, such a 3D printer could be used in a Virtual Reality or Augmented Reality application to dynamically form a tangible object or controller to provide haptic feedback and engage users physically. For children, it could dynamically form a physical educational manipulative, such as a molecular or architectural model, to learn and explore topics, for example in a science museum. Designers could use it to render a physical product to present to clients and interactively change the product's design through direct manipulation. In this vision, Dynamic 3D printing is an environment in which the user thinks, designs, explores, and communicates through dynamic and interactive physical representation.

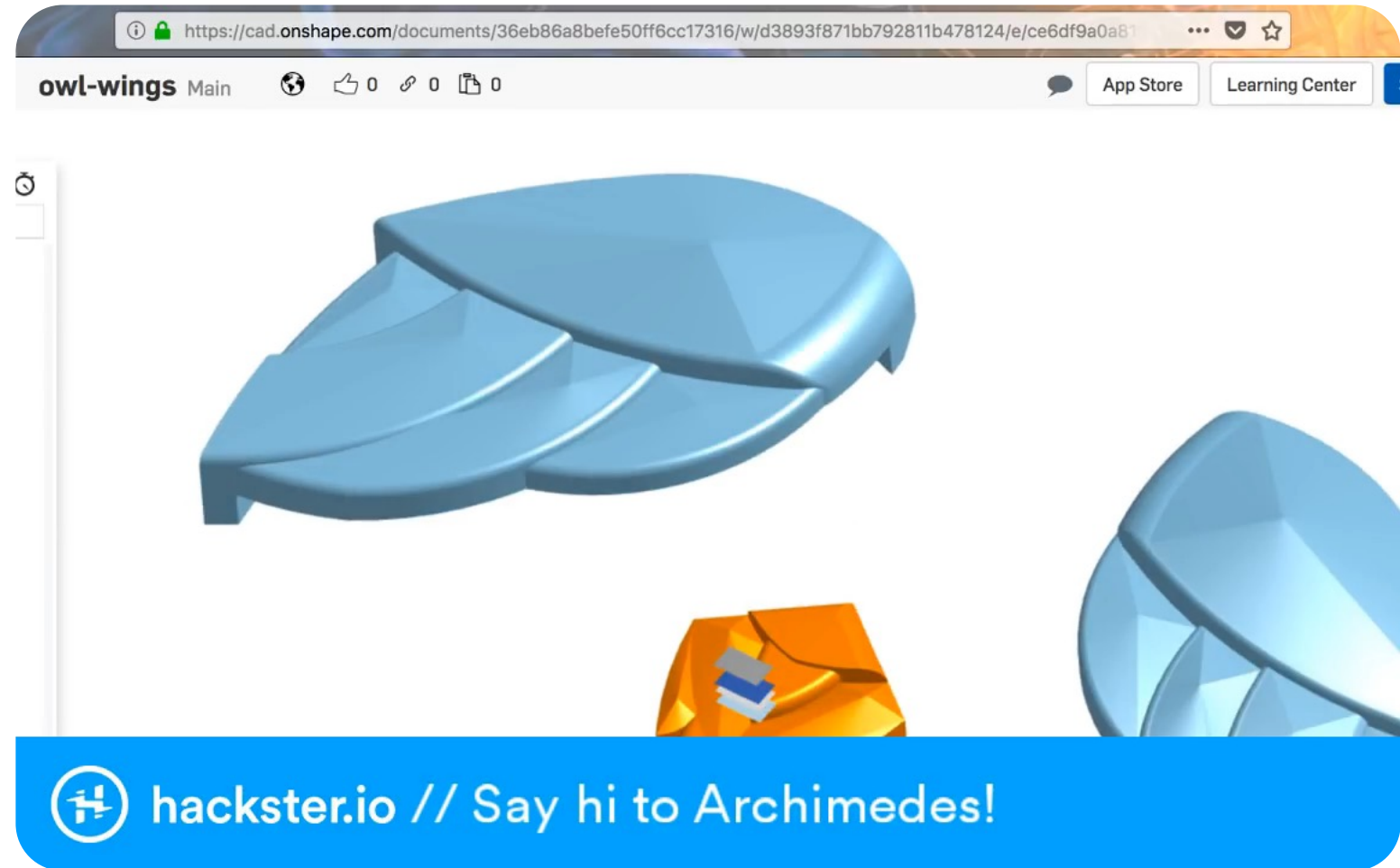
This paper develops this vision by proposing Dynamic 3D

UIST 2018

Suzuki et.al.

Possible topics

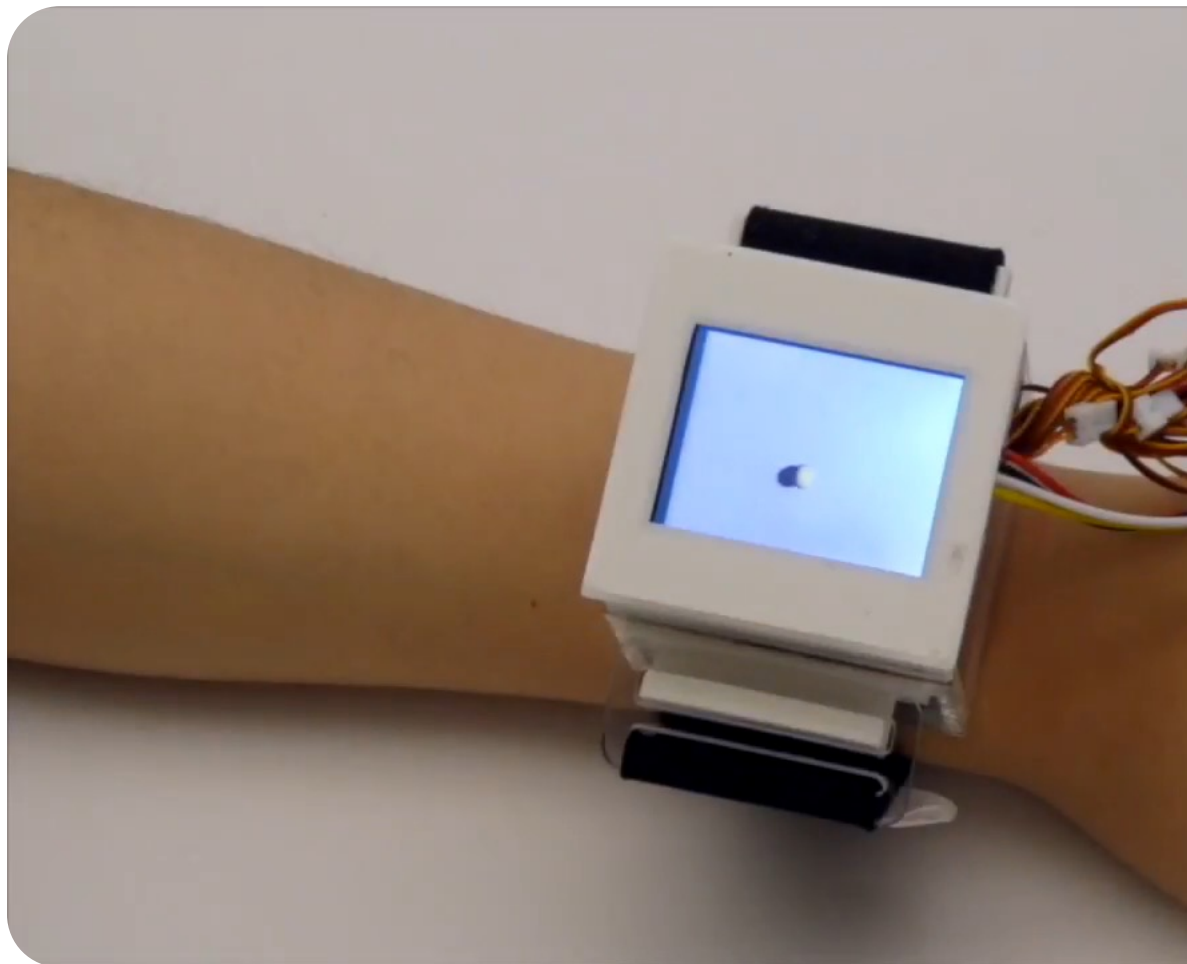
Wearables?



Archimedes: The AI Robot Owl.
Alex Glow.
2018

Possible topics

Wearables?



Session: Phones & Watches

UIST 2017, Oct. 22–25, 2017, Québec City, Canada

RetroShape: Leveraging Rear-Surface Shape Displays for 2.5D Interaction on Smartwatches

Da-Yuan Huang^{1,2}, Ruizhen Guo¹, Jun Gong¹, Jingxian Wang^{1,4}, John Graham¹,
De-Nian Yang³, Xing-Dong Yang¹

Dartmouth College¹, NTUST², Academia Sinica³, Carnegie Mellon University⁴
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dayuan.huang@csie.ntust.edu.tw, dnyang@iis.sinica.edu.tw, jingxian@cmu.edu

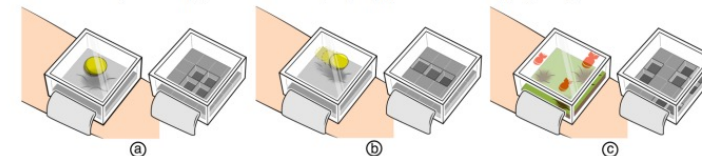


Figure 1. RetroShape aims to extend the visual scene to 2.5D physical space by a deformable display on its rear surface. Our RetroShape prototype equips 4×4 taxels, which can simulate (a) a bouncing ball on an elastic surface, (b) ball rolling, or (c) multiple strikes on the ground.

ABSTRACT

The small screen size of a smartwatch limits user experience when watching or interacting with media. We propose a supplementary tactile feedback system to enhance the user experience with a method unique to the smartwatch form factor. Our system has a deformable surface on the back of the watch face, allowing the visual scene on screen to extend into 2.5D physical space. This allows the user to watch and feel virtual objects, such as experiencing a ball bouncing against the wrist. We devised two controlled experiments to analyze the influence of tactile display resolution on the illusion of virtual object presence. Our first study revealed that on average, a taxel can render virtual objects between 70% and 138% of its own size without shattering the illusion. From the second study, we found visual and haptic feedback can be separated by 4.5mm to 16.2mm for the tested taxels. Based on the results, we developed a prototype (called RetroShape) with 4×4 10mm taxels using micro servo motors, and demonstrated its unique capability through a set of tactile-enhanced games and videos. A preliminary user evaluation showed that participants welcome RetroShape as a

Author Keywords

Mobile haptics; Shape-changing display; Taxel; Smartwatch

ACM Classification Keywords

H.5.2. [Information interfaces and presentation]: User Interfaces – Haptic I/O

INTRODUCTION

Smartwatches provide quick access to short-time entertainment applications, especially when users are on-the-move, e.g. in a bus or train. However, user experience in such applications is limited due to the small screen area and limited input and output options. While smartwatch visual and auditory technologies have improved substantially, the potential of smartwatch-enabled haptics in video and game applications remains to be exploited.

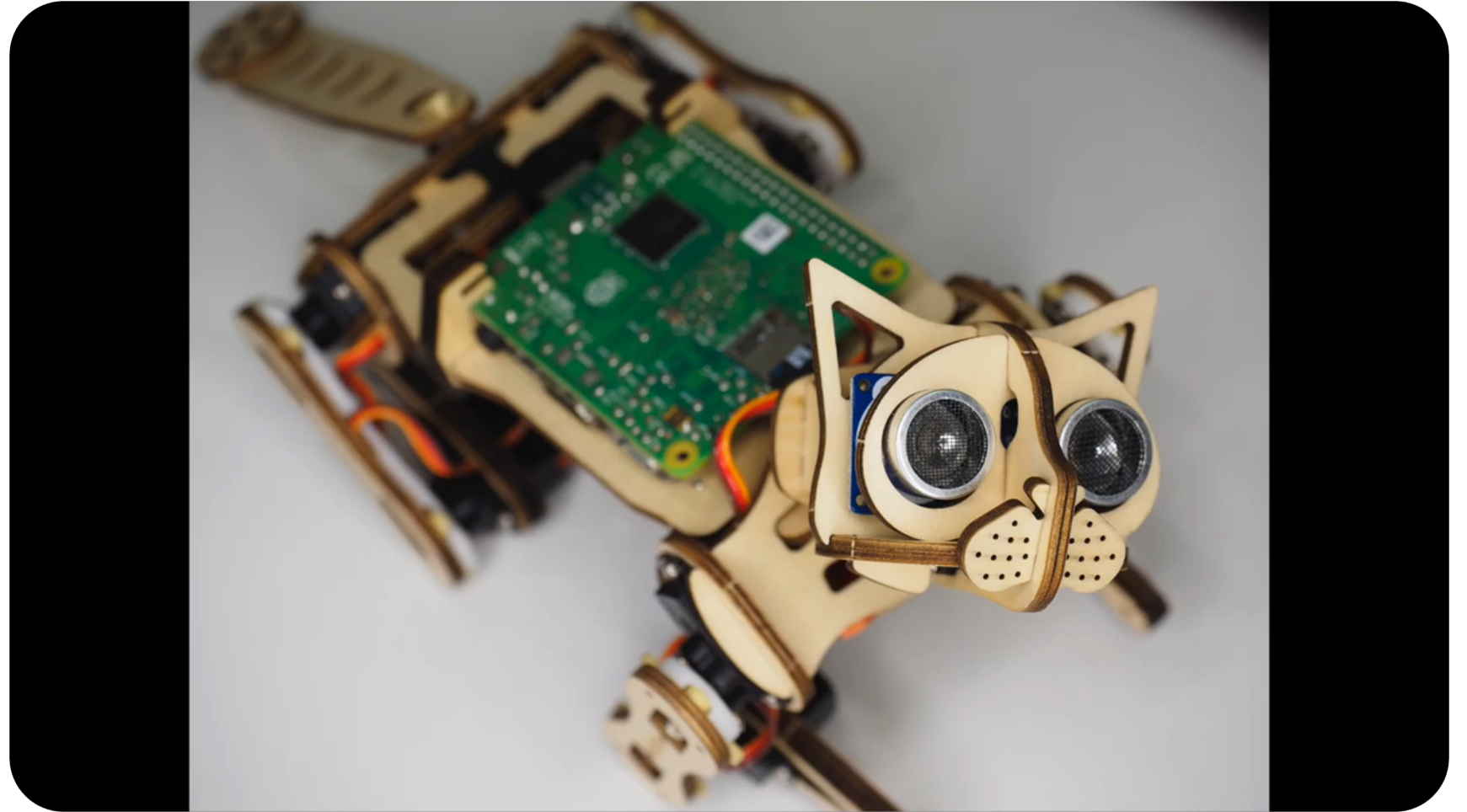
We leverage the user's skin under the watch face for sensing haptic output with collocated visual content. Our approach enhances the viewing experience on a smartwatch using a shape-changing tactile display on the rear surface of the smartwatch. Each pixel on the screen has a corresponding tactile pixel (or *taxel*) on the back of the watch face, allowing

UIST 2017

Huang et.al.

Possible topics

Desktop robot?

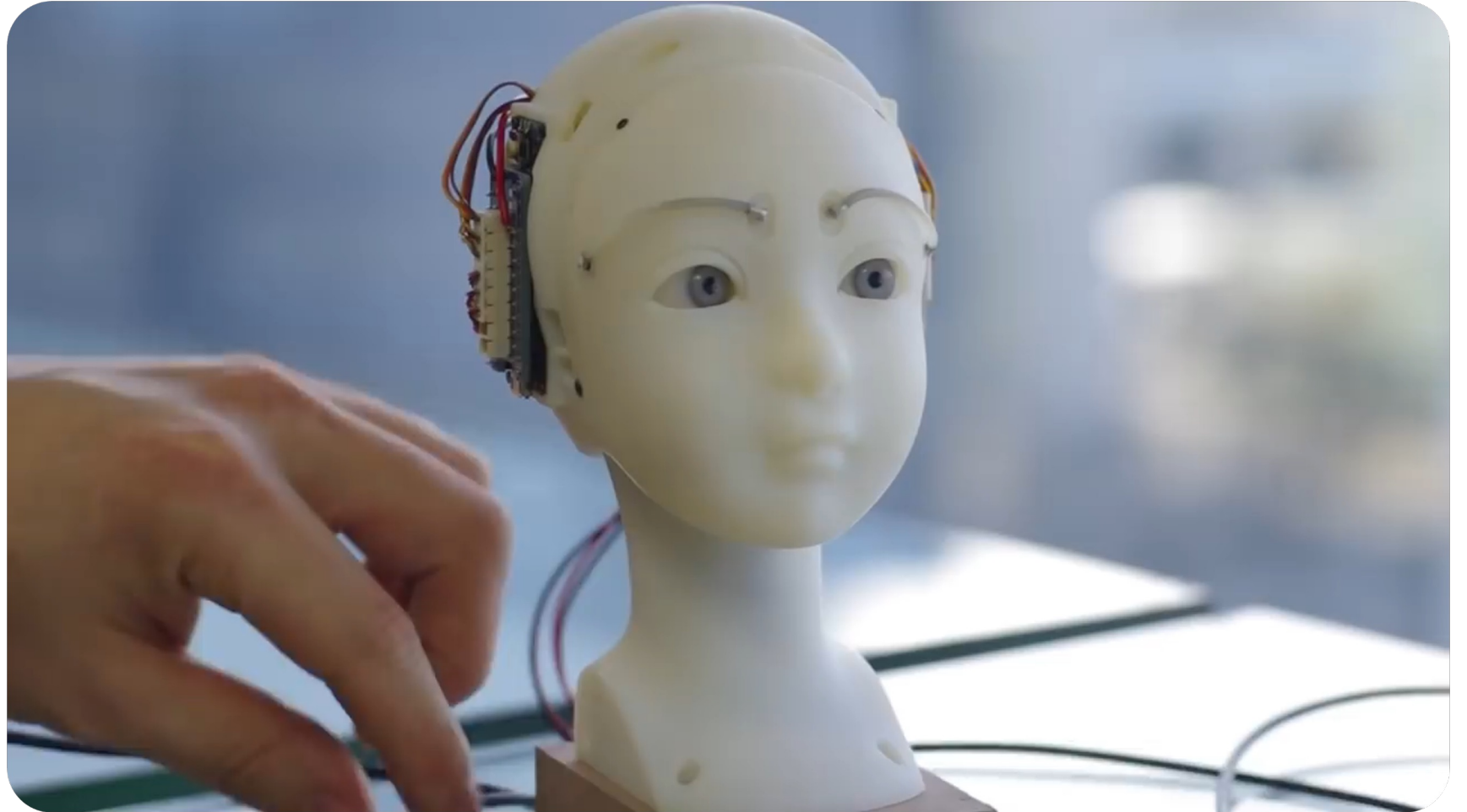


Raspberry Pi - Robot Cat Nybble

<https://www.youtube.com/watch?v=U2B1vQY1eKI>

Possible topics

Desktop robot?



By Takayuki Todo

<https://www.youtube.com/watch?v=BJZcGJSK1Z0&t>

From previous years...

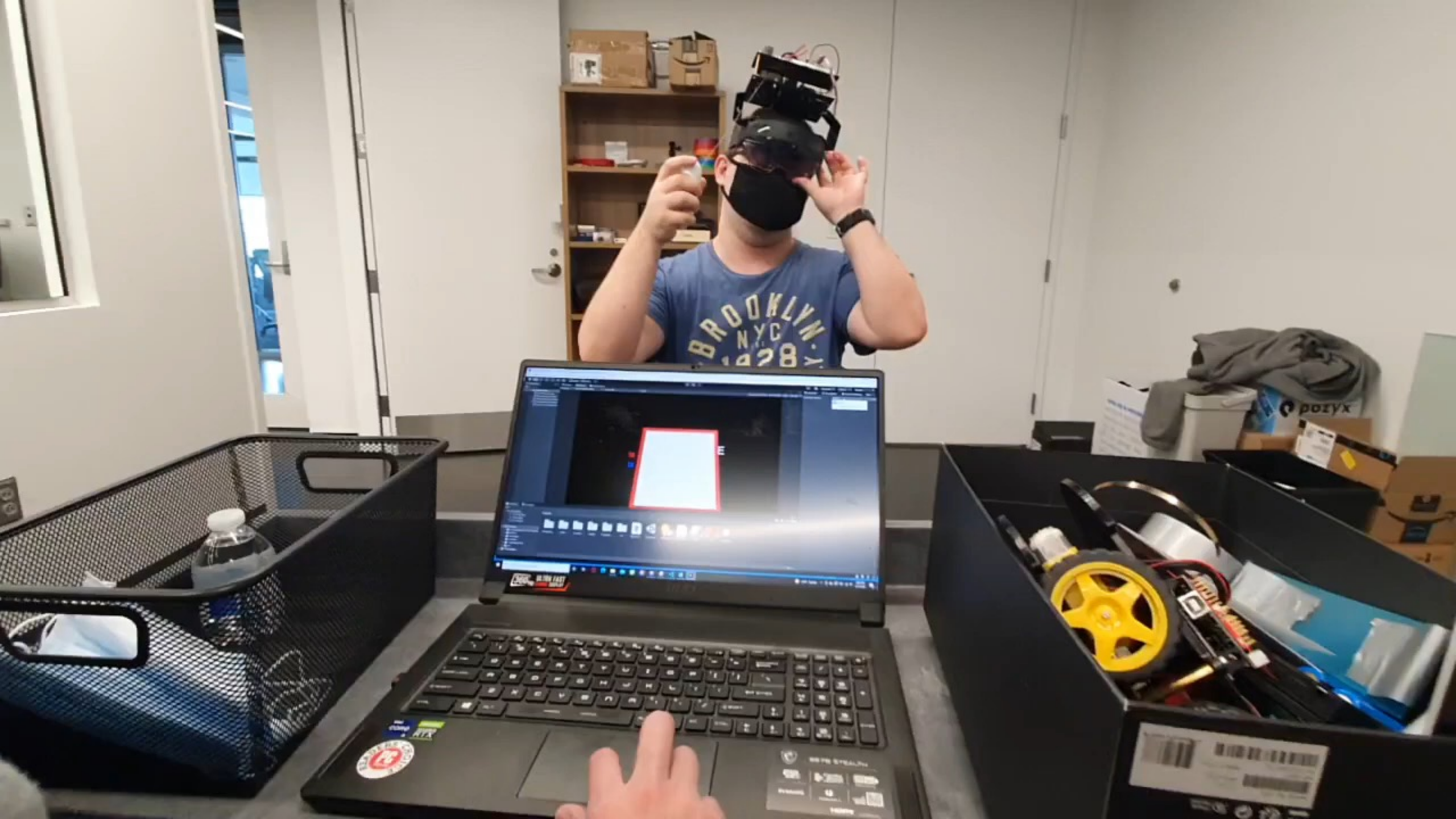
Drawing Beyond Touch: A Multi-user Collaborative Drawing System

Group 9

Biao Jia, Yue Jiang, Zehua Zeng, Yuheng Lu







MaSketch

**Andreas Verdelis, Kelsey Rassmann, Samuel Lam,
Srivatsan Srinivasan, and Sumbul Zehra**

More to be seen here:
<https://www.hackster.io/smartlab/projects>



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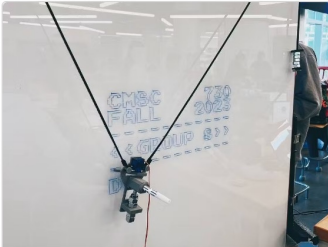
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Debuggy Ducky
Team Group 5

10

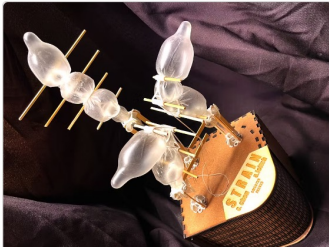
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Whiteboard drawing robot
Team Group 8 - Drawing Bot

15

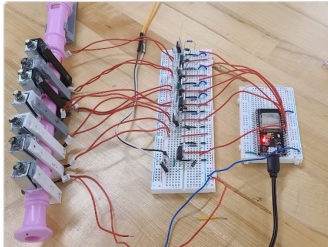
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Strain
Team CMSC730 Group 7

7

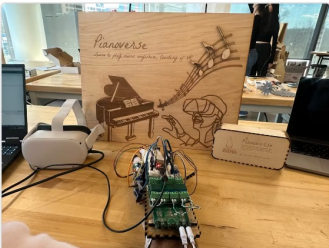
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Humming Robot
Team Humming Robot

8

300



Pianoverse

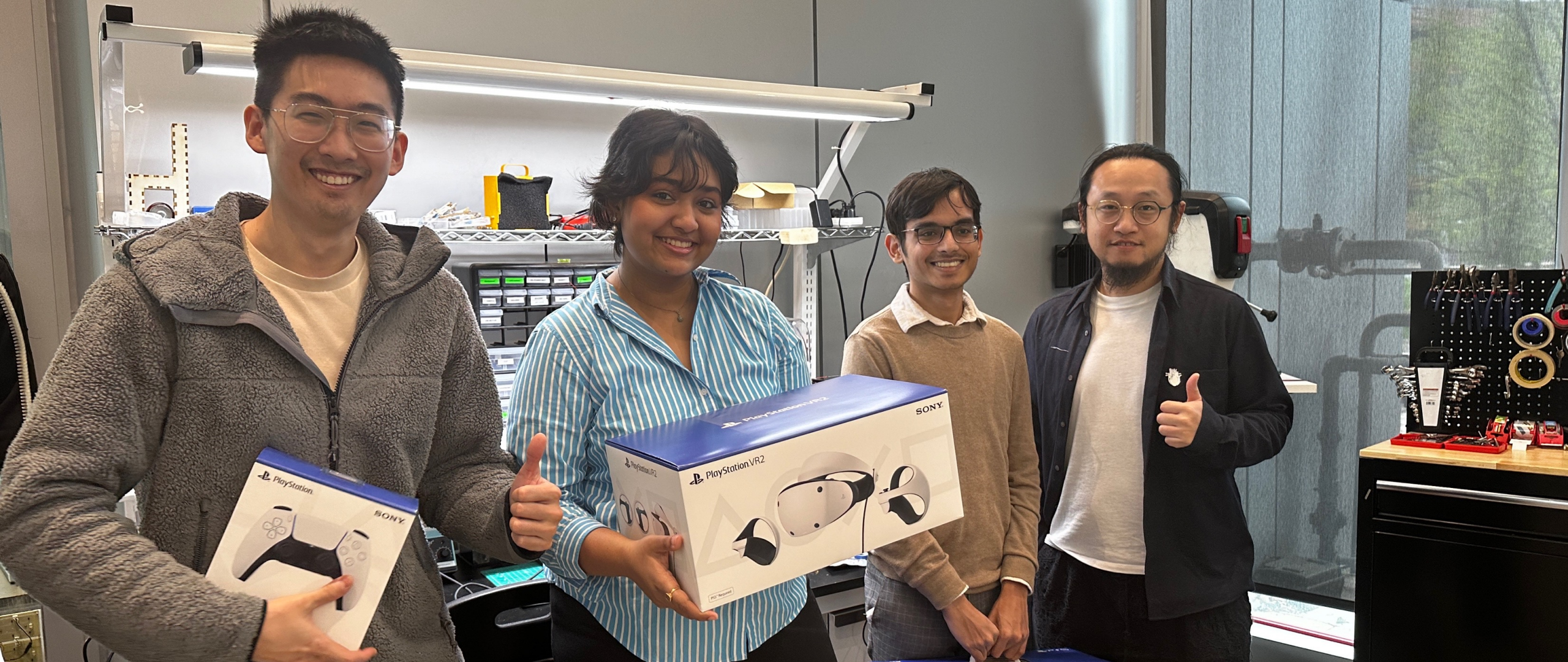


Pixar-Inspired Interactive Tracking Lamp



Musical Cocoon

Winners of Sony Spresense University Program 2023



Semester-long team project

Requirement:

- a) Has to be a robot, a tangible interface, or a haptic device
- b) Involve both **hardware** + **software** components
- c) Need to be **interactive**
- d) Not strict replication – **novelty (Combining with CV? LLM? Crowdsourcing?)**

We will cover basic techniques, but you are likely to **learn many new skills on your own.**

Don't be afraid of making mistakes,
TA and I are here to help

That's all very exciting but...

how much will it cost?

Do I need to pay a material fee?

Material fee

~~We charge you \$30 material fee~~

Material fee

We will provide you **\$30-worth** basic electronic components

Each **team** will also receive up-to **\$100** project budget (and you can recycle components from your individual parts)

Addition devices you can use from the SMART lab:



raspberry pi 5



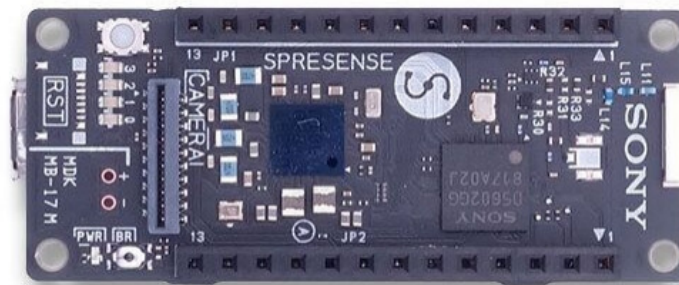
2 snapchat spectacles 4 AR glasses



1 4DOF basic robotic arm toy



360 Degree 12 Meters Scanning
Radius LIDAR



10 Sony Spresense Cam + Main Board

Addition devices you can use from the SMART lab:

If your team project requires something beyond the budget, talk to me.

I will purchase it if it can be reused for future classes.

If all sounds good ->

Declare your group by **Tuesday next week**

Find your peers here on ELMS Discussion forum

Class website

CMSC730

Home

Schedule

Robot Competition

Syllabus

Welcome to Spring 2026 CMSC730;

This is a graduate-level, research-oriented course covering broad areas of interactive technology and [human-computer interaction](#) (HCI) topics, *e.g.*, [ubiquitous computing](#), [wearables](#), [virtual/augmented reality](#), [haptics](#), [tangible UIs](#), [accessibility](#), and [Interactive fabrication](#).

The course consists of four modules. (1) **lectures**, through which we will examine major research topics in technical HCI; (2) **labs**, with which students will be equipped with a set of skills to make rapid and interactive physical prototypes, (3) **mini competition**, which require student dyad to use skills learned from the series of labs to design and build a rope climbing robot for a climbing competition, and (4) **semester-long project**, where students form a team of 3 to 4 and build a working prototype that solve one of the HCI/interation challenges.

Course Resources

ELMS Page	Contact TA	Contact Instructor
LOCATION AND TIME	Tu & Th, 3:30-4:45, IRB2207 + Sandbox	
INSTRUCTOR OFFICE HOURS	Th, 2:00-3:00, IRB4126 or Zoom	
TA OFFICE HOURS	Tu, 2:00-3:00, IRB0102	

Acknowledgment

CMSC730 is designed by [Huaishu Peng](#), who takes great inspiration from a suite of course materials designed by the following faculties: [Prof. François Guimbreti re](#) and [Prof. Cheng Zhang](#) @ Cornell, Prof. [Stefanie Mueller](#) @ MIT, Prof. [Jon Froehlich](#) @ UW, and Prof. [Bilge Mutlu](#) @ WISC. If you find an error please [contact me here](#).

<https://smartlab.cs.umd.edu/CMSC730/>

How you will be evaluated

Gradings	Category Weight
Assignments: 2D/3D drawing, laser cutting/3D printing, circuit, etc.	10%
Mini Robot Competition	15%
Semester-Long Project:	
Milestone 1	5%
Milestone 2	10%
Milestone 3	25%
Participation	5%
Final Exam	30%
100%	

Final exam (30)

Will cover basic topics from the course and it will be open-book

Small Assignments (10)

Help you go through the skills we taught in class

Will not take much of your time, but **you need to submit on time** (via ELMS)

You need to contact TA and CC me **at least 24h** before the deadline if you need an extension. Late assignment will be deducted 5% each day. See Late Assignments section in the Syllabus.

Participation (5)

Show up on time

Install required software before the class

Participate in discussion

Mini Robot Competition (15)

No need for detailed documentation. You will submit 3D drawing and schematics of your design. You will be graded based on the technical design and your robot performance.

Team Projects (5 + 10 + 25)

Milestone 1: Present 3 best project ideas + part list for the final decision

Milestone 2: First working prototype and demo/video at class

Milestone 3: Final working prototype + presentation

Enrollment

We will mainly use ELMS. The class website is a good one-place entry for everything.

ELMS for uploading assignments and for posting discussion questions, e.g., forming class teams.

Class location

Unless announced otherwise:

Tuesday: IRB 2206

Thursday: Sandbox

Resources

Course material will be given to you 2 weeks after the semester starts

All lectures will be recorded in case you run into difficulties

Office hours:

Tu/Tr 2:00-3:00 pm and by appointment –IRB0102/ IRB4126

Resources

LAB IRB 0102 – TA resource

IRB Sandbox (1st floor)



<https://sandbox.iri.be.umd.edu/>

For the rest of this class

Complete this online survey so that I can better know you and know what you would like to learn from this class (will not use for grading)

<https://tinyurl.com/cmssc730spring2026>

